Implementable Inventory Policies for Small-Scale Retailers with Limited Traceability Systems

Abstract. Problem definition: Motivated by a collaboration with a small-scale online grocer operating a large perishable assortment under severe traceability constraints, this paper studies periodic-review inventory management when the true on-hand stock is unobservable because hidden stochastic shrinkage—spoilage, theft, physical damage, and record error—continuously erodes usable inventory between replenishment cycles.

Methodology/results: We propose an algorithm that operates over a family of expected-inventory base-stock policies. At every stockout boundary, the algorithm faces a choice: launch a decision cycle to gather an evidence for a candidate base-stock level, or conduct a clean probe—a period in which a token replenishment order is placed and a complete manual count of the ending inventory is conducted. We establish a tight $\widetilde{O}(T^{2/3})$ regret upper bound against the best continuous base-stock level, with the manual-count schedule chosen to optimally balance the labor cost of counting against the shrinkage-inaccuracy cost of operating with an imprecise estimate.

Managerial implications: Our results reframe periodic inventory audits not as a fixed operational overhead but as an adaptive information-gathering investment whose frequency should be determined by the shrinkage characteristics of the product category being managed. Furthermore, we show that the inventory traceability that large retailers achieve through expensive ERP and RFID systems can be replicated by using nothing more than simple data infrastructure that small-scale retailers actually possess.

Key words: implementable inventory management; shrinkage; censored demand; online learning; base-stock policy

1. Introduction

Online retailing has fundamentally reshaped the competitive landscape of the retail industry. The dramatically reduced cost of entry into online channels has simultaneously empowered established players to scale further and enabled a new wave of small-scale entrepreneurs to launch viable retail businesses from scratch. Indeed, one of the most striking features of the past decade is that online retailing has benefited both ends of the size spectrum: large incumbents have extended their reach while small-scale operators have proliferated across virtually every product category, from fresh produce and specialty food to electronics, gardening equipment, and artisan goods. This democratizing effect has been amplified by the COVID-19 pandemic, which accelerated consumer adoption of online channels and created new openings for niche retailers able to supply products that were unavailable through mainstream brick-and-mortar stores.

Despite this rapid growth, the operations management literature has been slow to adapt to the inventory management challenges faced by small-scale entrepreneurs. The dominant paradigm in inventory management—built around dynamic programming models that treat the current inventory level as a state variable and update it after each period's realized sales—was developed primarily with large, resource-rich retailers in mind. These models assume, at their core, that the retailer can observe its inventory position accurately and in real time. This assumption is well justified for retailers with access to modern enterprise resource planning (ERP) systems, and sophisticated warehouse management software. For small-scale online retailers, however, it is rarely satisfied, and the gap between model assumptions and operational reality is large enough to render the standard framework inapplicable.

Small-scale online retailers typically operate with limited IT infrastructure. Many rely on basic spreadsheet tools or simple database systems that are updated manually rather than automatically. In such environments, the inventory record is perpetually at risk of diverging from the physical stock, because every manual update introduces opportunities for error or omission. More fundamentally, even if sales are recorded faithfully, the true inventory level is also eroded by forces that are largely unobservable: product spoilage, physical damage, misplacement, and theft—collectively referred to in the literature as *inventory shrinkage* or *scrappage*. Because scrappage events are rarely logged in real time, the retailer's recorded inventory tends to misrepresent what is actually available, a discrepancy that compounds over time.

One might argue that periodic counting can resolve this traceability problem. Under a periodic review policy, the retailer counts its stock just before placing a replenishment order, thereby resetting the inventory record to the true level at each review epoch. This argument, however, breaks down as the assortment grows. When a retailer manages hundreds or thousands of SKUs—a scale that is increasingly common even among small online operators—a complete manual count before each ordering decision becomes prohibitively time-consuming and costly. The practical consequence is that small-scale retailers often make replenishment decisions without knowing their true inventory levels, a constraint that existing dynamic inventory models do not accommodate.

This paper is motivated directly by our collaboration with one such retailer: a small-scale online grocer founded by an individual entrepreneur who identified an opportunity to connect local farmers and specialty food and beverage producers directly with end consumers in Eastern Canada. Leveraging online retailing capabilities and local supplier networks, the founder scaled the assortment from a few hundred SKUs to more than 10,000 SKUs in under three years. During this period of rapid growth, the COVID-19 pandemic provided an additional tailwind, as consumers sought fresh, locally sourced products that were difficult to obtain from large supermarket chains subject to supply-chain disruptions and capacity restrictions. As order volumes grew, the founder rented a local warehouse to consolidate inbound shipments from suppliers and fulfill outbound consumer orders.

The opening of a warehouse gave rise to a new challenge that had previously been manageable at small scale: the stock of perishable food and beverage products is subject to stochastic spoilage patterns that vary across suppliers, product types, and seasonal conditions. As a result, the warehouse's physical inventory diverges continuously from whatever counts exist in the retailer's simple database. The founder identified two ways to recover accurate inventory information. The first is a complete manual count—accurate but costly in labor and time, and therefore infeasible to perform at the frequency that good replenishment decisions would require. The second is to wait until the inventory of a given SKU is completely exhausted, at which point a stockout provides a reliable—though a destructive—signal that inventory has been depleted. The second approach, however, carries a direct cost in the form of lost sales and unserved demand. Neither option is operationally satisfactory, and neither is captured by standard dynamic inventory models.

Motivated by this setting, we take the perspective of resource-constrained small-scale online retailers and develop an inventory management framework that explicitly respects the traceability limitations they face in practice. We model the inventory level as an unobservable state variable whose evolution is driven by both realized sales—which are observable—and stochastic scrappage—which is not. The retailer therefore does not have access to the true inventory at the time of any given ordering decision. We develop an inventory policy adapted to this information structure, characterize its theoretical properties, and evaluate its performance relative to benchmarks.

Main Contributions. Our main results and contributions can be summarized as follows.

1. **A periodic-review inventory model with unobservable scrappage and lost sales.** We formulate a periodic-review inventory system in which the usable inventory available for sale is determined by a random survival factor—capturing spoilage, theft, damage, and record error—applied to on-hand stock before demand is realized. Because neither the scrappage factor nor lost demand is directly observable, the retailer's only signals are realized sales and whether a stockout occurred in each period. Motivated by our industry partner's practice, we model two mechanisms that create an exact resynchronization point with the true inventory: a *natural stockout*, in which the shelf empties and true inventory is known to be zero, and a *manual count*, in which the retailer conducts a full physical count of the warehouse.

The key asymmetry between the two mechanisms is informational: a natural stockout generates a *right-censored* observation of the scrappage factor (demand may have been satisfied before all shrinkage was realized), whereas a manual count yields a fully *uncensored* draw (the physical count reveals the exact surviving fraction). We show that the per-period true cost can be expressed, up to a single unknown nuisance parameter (the mean survival factor μ_ρ), entirely through two observable cycle-level aggregates: total sales and the difference between cumulative orders and cumulative sales over a completed cycle. This *pseudo-cost representation* (Proposition 1) is the structural foundation of the entire learning approach: it reduces an intractable partially observed control problem to a one-dimensional nuisance estimation problem coupled with a bandit-style search over base-stock levels.

2. **An implementable expected-inventory base-stock policy and a learning algorithm.** We restrict our attention to a one-parameter family of *expected-inventory base-stock policies* π_S^ν: rather than tracking the unobservable true inventory, each policy maintains an internal tracker $\hat{X}_t$ that is updated after each period using only the observable signals (Y_t, Z_t) and a tracker parameter ν approximating μ_ρ. At each stockout the tracker resets to zero exactly, matching the true state. We propose the **Kaplan-Meiyer (KM) Probe** algorithm, which learns the best policy in this class by interleaving two types of episodes at every stockout boundary: *decision cycles*, which run a candidate base-stock level and record the resulting cycle aggregates (V_k, W_k); and *clean probes*, which implement a complete manual count of the warehouse at the stockout boundary—the retailer conducts the physical count at the close of the stockout period, and thereby recovers an exact, uncensored draw of the scrappage realization at the cost of the counting labor rather than a loss of sales. All episode-start observations—from both decision cycles and manual counts—are pooled into a single unified censored-data log, from which three complementary confidence certificates for μ_ρ are maintained: a bracketing certificate (valid for any episode using deterministic bounds on the scrappage draw), a manual-count-only Hoeffding certificate (a robust fallback built from uncensored count observations), and a Kaplan–Meier certificate that exploits the full censoring structure of all episode-start observations and sharpens automatically when natural stockout observations are informative. The resulting confidence set drives both tracker implementation and pseudo-cost evaluation for arm comparison. Critically, the algorithm requires only the minimal data infrastructure that small-scale retailers actually possess: sales records and order quantities.

3. **Tight regret guarantees: a $\widetilde{O}(T^{2/3})$ upper bound and a matching lower bound.** We establish a $\widetilde{O}(T^{2/3})$ regret upper bound for the KM-Probe algorithm against the best continuous base-stock level $S^{\star}_{\mathrm{EV}}$ (Theorem 1). The bound makes the cost structure transparent: regret decomposes into a bandit-learning term from comparing arms via pseudo-cost estimates, and a shrinkage-learning term driven by cumulative uncertainty about μ_ρ. The manual-count schedule $a_m \asymp (L_T/m)^{1/3}$ is chosen to balance two opposing costs: the labor cost of conducting manual counts ($C_{\mathrm{pr}} P_T$, growing in the number of counts) and the shrinkage-inaccuracy cost incurred when replenishment cycles are run with an imprecise estimate of μ_ρ ($C_{\mathrm{sh,T}} \sum_m r_\mu(m)$, shrinking as counts accumulate). The schedule makes these two terms the same order, yielding the optimal $\widetilde{O}(T^{2/3})$ rate, and has a concrete operational interpretation: manual counts play the role of periodic inventory audits, and the algorithm determines their frequency adaptively depending on how quickly natural stockout observations already sharpen the shrinkage estimate. We complement this with a minimax lower bound of $\Omega(T^{2/3})$ proved on a deliberately simple hard subclass—deterministic demand, granular base-stock levels, and full reset after each period—showing that the $T^{2/3}$ rate is unavoidable and is driven by the fundamental information cost of learning hidden shrinkage (Theorem 2).
4. **Numerical experiments and managerial insights.**

   Mehmet: Numerical experiments are not yet complete. Text below should be expanded once experiments are finalized.

   We evaluate the proposed algorithm through numerical experiments and discuss the managerial implications of the theoretical findings.

Organization. The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model formulation. Section 4 develops the proposed ordering policy. Section 5 establishes the theoretical performance guarantees. Section 6 reports the numerical experiments and managerial insights. Section 7 concludes.

2. Literature

Our use of Kaplan–Meier inference is distinct from prior work in inventory learning and censored bandits. Huh et al. [2011] use the Kaplan–Meier estimator for online inventory learning with censored demand and establish convergence, while Lyu et al. [2024] use Kaplan–Meier recovery inside a UCB simulation procedure to estimate the demand distribution from censored sales and reduce counterfactual simulation bias.

In threshold bandits [Abernethy et al., 2016], KM-type estimators are used in finite-threshold censored-feedback problems, where observations at larger thresholds inform lower-threshold rewards. In contrast, our Kaplan–Meier component is applied to hidden shrinkage rather than demand, and the resulting confidence interval controls a global nuisance parameter that affects both policy evaluation and policy implementation. The statistical challenge is also different: in our setting the censoring is policy-dependent and non-identically distributed across episodes. We derive a finite-sample restricted-mean concentration certificate for the pooled product-limit estimator under this predictable censoring structure. This technical result may be of independent interest for adaptive inventory and censored-feedback learning problems.

3. Model

We study a periodic-review inventory system with hidden stochastic shrinkage and lost sales. The inventory available for sale is reduced by an unobserved random survival factor before demand arrives; the decision-maker observes only realized sales and whether a stockout occurred. Our goal is not to solve the full partially observed control problem. Instead, we restrict attention to a one-parameter family of expected-inventory base-stock policies that can be implemented from exactly these partial observations, and we study how to learn the best policy in this class.

3.1. True dynamics and observations

Time is indexed by $t = 1, 2, \ldots$. There is no lead time. Let

$$X_t \geq 0$$

denote the true on-hand inventory at the beginning of period t, before ordering. The controller places an order $q_t \geq 0$, so the pre-shrinkage inventory becomes $X_t + q_t$. A random survival factor

$$\rho_t \in [0, 1]$$

is then realized. We interpret ρ_t as the fraction of inventory that remains usable after spoilage, theft, breakage, record inaccuracy, or other shrinkage. The usable inventory available for sale in period t is therefore

$$A_t := \rho_t (X_t + q_t).$$

Demand in period t is denoted by $D_t \geq 0$. Sales are censored by usable inventory:

$$Y_t := \min\{A_t, D_t\}.$$

Lost demand and next-period true inventory are

$$L_t := D_t - Y_t = (D_t - A_t)^+, \qquad X_{t+1} := (A_t - D_t)^+.$$

The observable data in period t are

$$(Y_t, Z_t), \qquad Z_t := \mathbf{1}\{D_t > A_t\},$$

where Z_t is the stockout indicator. The true inventory X_t, the usable inventory A_t, the lost demand L_t, and the realized shrinkage factor ρ_t are unobserved.

We assume that $\{D_t\}_{t\geq 1}$ are i.i.d., $\{\rho_t\}_{t\geq 1}$ are i.i.d., and the two sequences are mutually independent. Their distributions are unknown to the learner. The mean survival factor is denoted by

$$\mu_\rho := \mathbb{E}[\rho_t].$$

The one-period cost is

$$C_t := hX_{t+1} + bL_t = h(\rho_t(X_t + q_t) - D_t)^+ + b(D_t - \rho_t(X_t + q_t))^+,$$

where $h > 0$ is the holding-cost parameter and $b > 0$ is the lost-sales penalty.

3.2. Expected-inventory base-stock policies

Because the true inventory is hidden, the policy class maintains an internal estimate of inventory rather than the physical state itself. For a tracker parameter $\nu \in (0, 1)$, let $\hat{X}_t$ denote the expected-inventory tracker. After ordering q_t and observing (Y_t, Z_t), the tracker is updated according to

$$\hat{X}_{t+1} = (1 - Z_t)\big(\nu(\hat{X}_t + q_t) - Y_t\big)^+. \tag{1}$$

The reset after a stockout is exact: if $Z_t = 1$, then all usable inventory has been sold, so both the true inventory and the tracker are zero at the next period.

For a base-stock level $S \in [0, \bar{S}]$, the expected-inventory base-stock policy with tracker parameter ν orders

$$q_t = (S - \hat{X}_t)^+. \tag{2}$$

We denote this policy by π_S^ν. The benchmark policy class corresponds to the physically correct tracker parameter $\nu = \mu_\rho$. Thus the benchmark policy indexed by S is $\pi_S^{\mu_\rho}$, even though μ_ρ is unknown to the learner.

For any $S \in [0, \bar{S}]$ and $\nu \in (0,1)$, define the long-run average cost

$$g(S;\nu) := \limsup_{T\to\infty} \frac{1}{T} \mathbb{E}^{\pi_S^{\nu}} \left[\sum_{t=1}^{T} C_t \right]. \tag{3}$$

We write

$$g(S) := g(S; \mu_\rho)$$

for the cost of the policy in the intended expected-inventory class. The best policy in this class is

$$S^{\star}_{\mathrm{EV}} \in \arg\min_{S\in[0,\bar{S}]} g(S). \tag{4}$$

For an online algorithm, regret over a horizon of T periods is measured against this continuous benchmark:

$$R_T := \mathbb{E}\left[\sum_{t=1}^{T} C_t^{\mathrm{alg}} \right] - T g(S^{\star}_{\mathrm{EV}}). \tag{5}$$

The hidden-shrinkage setting differs from standard censored-demand learning in one crucial respect. Estimating μ_ρ is not needed only to compare policies. It is also needed to *implement* them, because the tracker dynamics in (1) depend directly on the chosen value of ν. If the learner uses $\nu \neq \mu_\rho$, then even at the same base-stock level S the realized order sequence differs from that of the benchmark policy.

3.3. Stockout cycles and observable pseudo-costs

The true period cost is not observable. The learner observes sales and stockout indicators, but not the next true inventory X_{t+1}, lost demand L_t, usable inventory A_t, or the shrinkage realization ρ_t. The key observation is that stockout cycles nevertheless contain enough observable aggregate information to evaluate policies up to a one-dimensional nuisance parameter.

Stockouts create regeneration points. Define the stockout times recursively by

$$\tau_0 := 0, \qquad \tau_k := \inf\{t > \tau_{k-1} : Z_t = 1\}, \qquad k \geq 1,$$

and let

$$\mathcal{C}_k := \{\tau_{k-1}+1, \ldots, \tau_k\}$$

be the kth completed stockout cycle. For a completed cycle, define its length, total replenishment, total sales, and observable inventory loss by

$$\Lambda_k := \tau_k - \tau_{k-1}, \qquad Q_k := \sum_{t\in\mathcal{C}_k} q_t, \qquad V_k := \sum_{t\in\mathcal{C}_k} Y_t, \qquad W_k := Q_k - V_k.$$

The quantity W_k is observable because it is the difference between total orders and total sales. It also has a physical interpretation: it is exactly the total inventory lost to shrinkage over the completed cycle. Indeed, the inventory balance can be written as

$$X_{t+1} = X_t + q_t - (1-\rho_t)(X_t + q_t) - Y_t.$$

Summing over a completed cycle, and using that both the initial and terminal true inventories are zero, gives

$$Q_k = V_k + \sum_{t \in \mathcal{C}_k} (1-\rho_t)(X_t + q_t). \tag{6}$$

Hence

$$W_k = \sum_{t \in \mathcal{C}_k} (1-\rho_t)(X_t + q_t).$$

We now derive the pseudo-cost. In every period,

$$X_{t+1} = A_t - Y_t, \qquad L_t = D_t - Y_t,$$

so the true cost can be rewritten as

$$C_t = h(A_t - Y_t) + b(D_t - Y_t) = bD_t + hA_t - (h+b)Y_t. \tag{7}$$

The term Y_t is observed, while D_t and A_t are not. Over a cycle generated by a fixed policy π_S^{ν}, the demand term contributes only the arm-independent constant $b\mathbb{E}[D_t]$ per period after normalization. The remaining challenge is to express the expected usable-inventory term in (7) through observable cycle aggregates.

For a fixed base-stock level S and tracker parameter ν, write

$$(\Lambda_k^{S,\nu}, Q_k^{S,\nu}, V_k^{S,\nu}, W_k^{S,\nu})$$

for the kth stockout cycle generated by π_S^{ν} from a stockout state. These are population cycle random variables under the fixed policy (S,ν). Since ρ_t is independent of the pre-shrinkage inventory $X_t + q_t$ and $\mathbb{E}[\rho_t] = \mu_\rho$, we have

$$\mathbb{E}\left[\sum_{t \in \mathcal{C}_1} \rho_t (X_t + q_t)\right] = \frac{\mu_\rho}{1-\mu_\rho} \mathbb{E}\left[\sum_{t \in \mathcal{C}_1} (1-\rho_t)(X_t + q_t)\right].$$

Using (6), this becomes

$$\mathbb{E}\left[\sum_{t \in \mathcal{C}_1} A_t\right] = \frac{\mu_\rho}{1-\mu_\rho} \mathbb{E}[W_1^{S,\nu}]. \tag{8}$$

For $\mu \in (0,1)$, define

$$\alpha(\mu) := \frac{h\mu}{1-\mu}. \tag{9}$$

Thus, the corresponding cycle pseudo-cost random variable is

$$\widetilde{K}_k^{S,\nu}(\mu) := \alpha(\mu) W_k^{S,\nu} - (h+b) V_k^{S,\nu}. \tag{10}$$

When the pseudo-cost coefficient is the true one, define the population pseudo-cost rate

$$\gamma(S;\nu) := \frac{\mathbb{E}[\widetilde{K}_1^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda_1^{S,\nu}]}. \tag{11}$$

In particular, $\gamma(S;\mu_\rho)$ is the benchmark pseudo-cost rate for base-stock level S.

The next proposition explains why this quantity is central.

Proposition 1 (Cycle pseudo-cost representation with observable sufficient statistics) *Fix $S \in [0,\bar{S}]$ and a tracker parameter $\nu \in (0,1)$ for which the induced stockout cycles are positive recurrent. Let*

$$v(S;\nu) := \frac{\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}, \qquad w(S;\nu) := \frac{\mathbb{E}[W_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}.$$

Then

$$g(S;\nu) = b\mathbb{E}[D_t] + \alpha(\mu_\rho) w(S;\nu) - (h+b) v(S;\nu) = b\mathbb{E}[D_t] + \gamma(S;\nu). \tag{12}$$

In particular, for the benchmark tracker $\nu = \mu_\rho$, minimizing $g(S)$ over S is equivalent to minimizing

$$\gamma(S;\mu_\rho).$$

Proposition 1 is the structural reduction on which the learning problem rests. The true period cost C_t is hidden because lost demand and remaining true inventory are hidden. A completed cycle, however, reveals the aggregate statistics V_k and W_k. The oracle pseudo-cost of that cycle is still not directly known because its coefficient $\alpha(\mu_\rho)$ depends on the unknown mean survival factor. Thus the learning problem separates into two coupled tasks: estimate the global nuisance parameter μ_ρ, and use realized cycle records to learn which base-stock level has the smallest population pseudo-cost rate.

We have the following assumptions.

Assumption 1 (Bounded demand) *There are known constants $0 < \underline{D} \le \overline{D} < \infty$ such that $\underline{D} \le D_t \le \overline{D}$ almost surely.*

Assumption 2 (Compact mean-shrinkage interval) *The unknown mean survival factor satisfies* $\mu_\rho \in [\underline{\mu}, \bar{\mu}] \subset (0,1)$.

Assumption 3 (Stockout-forcing shrinkage event) *There exists* $p_0 \in (0,1]$ *such that* $\mathbb{P}(\rho_t = 0) \geq p_0$.

Assumption 4 (Demand anti-concentration) *There exists a constant* $\bar{f}_D < \infty$ *such that for every interval* $I \subset \mathbb{R}_+$, $\mathbb{P}(D_t \in I) \leq \bar{f}_D |I|$, *where* $|I|$ *denotes the interval length.*

Assumptions 1–2 are boundedness conditions. Assumption 3 is the primitive recurrence condition that gives uniform control of stockout cycles. Assumption 4 controls the discontinuity created by stockout indicators when the tracker parameter or the base-stock level changes.

4. Proposed Algorithm: Kaplan-Meiyer (KM) Probe

The proposed algorithm is built around the regenerative structure created by stockouts. Each stockout boundary is a decision epoch. At that epoch the algorithm either starts a decision cycle by selecting a base-stock level from a finite grid, or conducts a complete manual inventory count—a *clean probe*—to obtain an exact, uncensored shrinkage observation and sharpen inference on the mean survival factor~~launches a deliberately small clean probe to sharpen inference on the mean survival factor~~[MG]. The central design principle is that *all* first-period episode observations—natural cycle starts and clean probes alike—are pooled into one unified censored-data problem for shrinkage learning. Ordinary decision cycles contribute right-censored observations of ρ_t; clean probes contribute uncensored ones. This unified sample drives a single confidence interval for μ_ρ, which is then fed back both into policy implementation and into pseudo-cost evaluation.

4.1. Finite grid and episode structure

The algorithm discretizes the base stock level. Fix

$$\mathcal{S}_M := \{s_1, \ldots, s_M\} \subset (0, \bar{S}].$$

The algorithm learns over this finite candidate set while regret is still measured against the continuous oracle $S^\star_{\mathrm{EV}}$ defined in (4). We also fix the finite Kaplan–Meier truncation grid

$$\mathcal{A}_T := \left\{0, \frac{1}{T}, \frac{2}{T}, \ldots, 1\right\}.$$

The episode-wise confidence budget used by the certificates is

$$\delta_m := \frac{1}{64T^4(m+1)^2}, \qquad m \geq 0.$$

Operationally, the algorithm proceeds in *episodes*. Episode m begins immediately after a stockout, at a time t_m when both the true inventory and the tracker are zero:

$$X_{t_m} = 0, \qquad \hat{X}_{t_m} = 0.$$

At this boundary the algorithm chooses an episode-start order

$$\xi_m \in \mathcal{S}_M \cup \{s_{\text{pr}}\}.$$

If $\xi_m = s_i \in \mathcal{S}_M$, episode m is a decision cycle for arm i: the algorithm starts the expected-inventory base-stock policy at level s_i and continues it until the next stockout. If $\xi_m = s_{\text{pr}}$, episode m is a *clean probe*: the retailer places a token replenishment order s_{pr} at the stockout boundary, operates normally for one period, and then conducts a complete manual count of the remaining on-shelf inventory X_{t_m+1} at the period's end. Because the true inventory at the boundary was zero, the usable inventory entering the period is $\rho_{t_m} s_{\text{pr}}$, of which Y_{t_m} units are sold and $X_{t_m+1} = (\rho_{t_m} s_{\text{pr}} - D_{t_m})^+$ units remain. Both Y_{t_m} and X_{t_m+1} are observed—the former from sales records and the latter from the physical count—so the exact shrinkage draw is recovered as $\rho_{t_m} = (Y_{t_m} + X_{t_m+1})/s_{\text{pr}}$, regardless of whether a stockout occurred during the period. The probe lasts one period by construction. Its operational cost is the labor and time required for the manual count, and is modeled as C_{pr} in the regret analysis (Section 4.4).~~If $\xi_m = s_{\text{pr}}$, episode m is a clean probe and lasts one period by construction.~~MG

Because $X_{t_m} = 0$, the episode-start order ξ_m is also the pre-shrinkage inventory in the first period of the episode. Therefore

$$Y_{t_m} = \min\{\rho_{t_m} \xi_m, D_{t_m}\}.$$

Define the normalized first-period observation

$$U_m := \frac{Y_{t_m}}{\xi_m}, \qquad C_m := \frac{D_{t_m}}{\xi_m}. \tag{13}$$

Then

$$U_m = \min\{\rho_{t_m}, C_m\}. \tag{14}$$

If $Z_{t_m} = 1$, then demand exceeds usable inventory and the shrinkage realization is revealed exactly:

$$U_m = \rho_{t_m}.$$

If $Z_{t_m} = 0$, then sales are demand-censored and

$$U_m = C_m \le \rho_{t_m}.$$

Thus each decision-cycle[MG] episode contributes a right-censored observation of the latent shrinkage draw. Manual-count (probe) episodes, by contrast, contribute exact uncensored draws via the inventory-balance recovery described below.~~. This is true for both decision cycles and probes.~~[MG]

The clean probe uses a token replenishment order $s_{\text{pr}} > 0$ of any size.~~The clean probe is the special episode-start order~~[MG] ~~$0 < s_{\text{pr}} < \underline{D}$.~~[MG] Unlike decision cycles, a probe does not rely on a stockout to reveal shrinkage information. Instead, the retailer conducts a physical count of the remaining inventory X_{t_m+1} at the close of the probe period. Since the true inventory at the boundary is zero, the inventory balance for the probe period reads[MG]

$$\rho_{t_m} s_{\text{pr}} = Y_{t_m} + X_{t_m+1},$$

where Y_{t_m} is observed from sales records and X_{t_m+1} is obtained from the physical count. Dividing both sides by s_{pr} recovers the exact shrinkage draw:[MG]

$$\rho_{t_m} = \frac{Y_{t_m} + X_{t_m+1}}{s_{\text{pr}}}.$$

Mehmet: The old forced-stockout argument has been fully replaced by the inventory-balance recovery argument above and is removed throughout this section. Does it make sense?

This recovery is exact regardless of whether a stockout occurred during the probe period: the physical count directly eliminates the censoring that would otherwise prevent observation of ρ_{t_m}. The probe observation entered into the unified shrinkage log is therefore the pair $(\rho_{t_m}, 1)$, indicating an uncensored draw.[MG] A clean probe is therefore not a separate estimator. It is a complete, uncensored observation in the same shrinkage sample used throughout the algorithm, obtained via the physical count without any restriction on the size of the token order~~obtained at the cost of a manual count rather than at the cost of a forced stockout~~[MG].

The unified shrinkage log after m completed episodes is

$$\mathscr{D}_\rho(m) := \{(U_\ell, Z_{t_\ell}) : 1 \le \ell \le m\}.$$

All information about μ_ρ is extracted from this single log.

4.2. Shrinkage learning from a unified censored log

The unified log $\mathscr{D}_\rho(m)$ is the sole source of information about the mean survival factor. It is useful because every episode-start observation has the same structure: the latent draw ρ_{t_m} is observed either exactly or through a right-censoring threshold. The algorithm exploits this structure at three levels.

The first level is purely order-theoretic and does not rely on any survival-analysis machinery. From (14) we always have

$$U_m \le \rho_{t_m} \le 1,$$

and when $Z_{t_m} = 1$ this bracket sharpens to the identity $U_m = \rho_{t_m}$. Thus every episode contributes a deterministic lower bound on the latent shrinkage draw, and uncensored episodes contribute exact draws. Averaging these lower and upper brackets yields a robust certificate for μ_ρ that is valid without any overlap requirement.

The second level uses the manual-count (clean probe) observations~~second level uses the clean probes~~MG. Because a manual count always reveals the shrinkage draw exactly, the probe subsequence is an i.i.d. sample in $[0,1]$ with mean μ_ρ~~a probe always reveals the shrinkage draw exactly, the probe subsequence is an i.i.d. sample in $[0,1]$ with mean μ_ρ~~MG. The resulting Hoeffding interval is statistically crude but extremely reliable. It is this certificate that underwrites the paper's robust upper bound when natural overlap is weak.

The third level uses the full censoring structure of the unified log through the Kaplan–Meier product-limit estimator. This is the statistically efficient component: when the realized censoring thresholds leave enough of the upper tail exposed, the Kaplan–Meier interval can shrink substantially faster than the probe-only certificate. The algorithm therefore treats manual-count probes ~~clean probes~~MG not as a different estimator, but as particularly informative uncensored observations inside the same censored-data problem.

We now define the three certificates used in the paper.

Bracketing certificate. For each observation define

$$L_\ell := U_\ell, \qquad H_\ell := \begin{cases} U_\ell, & Z_{t_\ell} = 1, \\ 1, & Z_{t_\ell} = 0. \end{cases}$$

Then $L_\ell \le \rho_{t_\ell} \le H_\ell$ almost surely. Let

$$\bar{L}_m := \frac{1}{m}\sum_{\ell=1}^{m} L_\ell, \qquad \bar{H}_m := \frac{1}{m}\sum_{\ell=1}^{m} H_\ell.$$

The bracketing certificate is

$$\mathcal{C}_{\mathrm{br}}(m) := \left[\bar{L}_m - r_{\mathrm{br}}(m), \bar{H}_m + r_{\mathrm{br}}(m)\right] \cap [\underline{\mu}, \bar{\mu}], \qquad r_{\mathrm{br}}(m) := \sqrt{\frac{\log(6/\delta_m)}{2m}}. \tag{15}$$

This interval is deliberately conservative: it ignores the precise censoring structure and uses only the deterministic fact that each observation brackets the latent shrinkage draw.

Manual-count (probe) certificate~~Probe certificate~~MG. Let $\mathcal{P}(m) \subseteq \{1, \ldots, m\}$ denote the manual-count (probe)~~probe~~MG episodes completed up to time m, and let $N_{\mathrm{pr}}(m) := |\mathcal{P}(m)|$. Because every manual-count~~probe~~MG observation is uncensored, the manual-count~~probe~~MG subsequence is an ordinary i.i.d. sample from the shrinkage distribution. Its Hoeffding certificate is

$$\mathcal{C}_{\mathrm{pr}}(m) := [\hat{\mu}_{\mathrm{pr}}(m) - r_{\mathrm{pr}}(m), \hat{\mu}_{\mathrm{pr}}(m) + r_{\mathrm{pr}}(m)] \cap [\underline{\mu}, \bar{\mu}], \tag{16}$$

where

$$\hat{\mu}_{\mathrm{pr}}(m) := \frac{1}{N_{\mathrm{pr}}(m)} \sum_{\ell \in \mathcal{P}(m)} U_\ell, \qquad r_{\mathrm{pr}}(m) := \sqrt{\frac{\log(6/\delta_m)}{2N_{\mathrm{pr}}(m)}}.$$

If $N_{\mathrm{pr}}(m) = 0$, we set $\mathcal{C}_{\mathrm{pr}}(m) := [\underline{\mu}, \bar{\mu}]$. This certificate is the robust fallback used in the main theorem.

Kaplan–Meier certificate. The Kaplan–Meier component is the overlap-adaptive part of the shrinkage confidence set. The bracketing and probe certificates are robust safeguards: they remain valid even when the natural episode-start observations are weakly informative. The Kaplan–Meier certificate uses the full censoring structure of the unified log and can produce substantially sharper intervals when the realized censoring thresholds leave enough of the shrinkage distribution exposed.

Although the censoring threshold

$$C_\ell = \frac{D_{t_\ell}}{\xi_\ell}$$

depends on the policy through the predictable episode-start order ξ_ℓ, this does not invalidate the pooled product-limit estimator. Conditional on the episode-start history, ξ_ℓ is fixed, while D_{t_ℓ} and ρ_{t_ℓ} are fresh and independent. Thus the censoring threshold is predictable and conditionally independent of the latent shrinkage draw. The censoring distribution may vary across episodes, but the latent shrinkage distribution is common. The pooled risk-set martingale therefore has the same shrinkage hazard in every episode.

Demand anti-concentration rules out censoring ties

$$\rho_{t_\ell} = D_{t_\ell}/\xi_\ell,$$

but it does not rule out ties among shrinkage realizations. Those ties are allowed. The product-limit estimator below uses grouped event counts and hence accommodates atoms of the shrinkage distribution, including the stockout-forcing atom in Assumption 3.

Let

$$0 \le r_{1,m} < \cdots < r_{J_m,m} \le 1$$

be the distinct uncensored values among

$$\{U_\ell : Z_{t_\ell} = 1,\ 1 \le \ell \le m\}.$$

For each such value define the grouped event count and risk set

$$d_{j,m} := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell = r_{j,m},\ Z_{t_\ell} = 1\}, \qquad n_{j,m} := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \ge r_{j,m}\}.$$

The Kaplan–Meier estimator of the survival function

$$\bar{F}_\rho(x) := \mathbb{P}(\rho_t > x)$$

is

$$\widehat{\bar{F}}_{\rho,m}(x) := \prod_{j: r_{j,m} \le x} \left(1 - \frac{d_{j,m}}{n_{j,m}}\right), \qquad x \in [0,1], \tag{17}$$

with the convention that an empty product equals one. For a truncation level $a \in [0,1]$, define the restricted-mean estimator

$$\widehat{\mu}_{\mathrm{KM}}(m; a) := \int_0^a \widehat{\bar{F}}_{\rho,m}(x)\, dx. \tag{18}$$

For $a \in \mathcal{A}_T$, define the empirical risk set at level a by

$$R_m(a) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \ge a\},$$

and define the product-limit cumulative event score

$$\widehat{\mathsf{H}}_m(a) := \sum_{j: r_{j,m} \le a} \frac{d_{j,m}}{n_{j,m}}.$$

Let

$$\ell_m^{\mathrm{KM}} := \log\left(\frac{32(m \vee 1)|\mathcal{A}_T|}{\delta_m}\right),$$

and set

$$\zeta_m(a) := 2\sqrt{\frac{(\widehat{\mathsf{H}}_m(a) + 1)\ell_m^{\mathrm{KM}}}{R_m(a) \vee 1}} + 8\frac{\ell_m^{\mathrm{KM}}}{R_m(a) \vee 1}. \tag{19}$$

The restricted-mean radius is

$$e_{\mathrm{KM}}(m;a) := \min\Big\{1,\, \exp\big(\widehat{\mathsf{H}}_m(a) + \zeta_m(a) + 1\big)\zeta_m(a)\Big\}. \tag{20}$$

The truncation-protected KM certificate is

$$\mathcal{C}_{\mathrm{KM}}(m;a) := [\widehat{\mu}_{\mathrm{KM}}(m;a) - e_{\mathrm{KM}}(m;a),\, \widehat{\mu}_{\mathrm{KM}}(m;a) + e_{\mathrm{KM}}(m;a) + (1-a)] \cap [\underline{\mu}, \bar{\mu}]. \tag{21}$$

The additive term $1-a$ is deterministic tail protection:

$$0 \le \int_a^1 \mathbb{P}(\rho_t > x)\, dx \le 1 - a.$$

The algorithm chooses the truncation level from the finite grid by

$$a_m^{\mathrm{KM}} \in \arg\min_{a \in \mathcal{A}_T} \mathrm{diam}\big(\mathcal{C}_{\mathrm{KM}}(m;a)\big), \qquad \mathcal{C}_{\mathrm{KM}}(m) := \mathcal{C}_{\mathrm{KM}}(m; a_m^{\mathrm{KM}}). \tag{22}$$

The shrinkage confidence set used by the algorithm is the intersection

$$\mathcal{C}_\mu(m) := \mathcal{C}_{\mathrm{KM}}(m) \cap \mathcal{C}_{\mathrm{br}}(m) \cap \mathcal{C}_{\mathrm{pr}}(m). \tag{23}$$

On the simultaneous confidence event, all three intervals contain μ_ρ, and hence so does their intersection. We summarize this interval by its midpoint and radius,

$$\hat{\mu}_m := \mathrm{mid}\big(\mathcal{C}_\mu(m)\big), \qquad r_\mu(m) := \frac{1}{2}\,\mathrm{diam}\big(\mathcal{C}_\mu(m)\big). \tag{24}$$

The tracker parameter used throughout episode $m+1$ is

$$\nu_m := \hat{\mu}_m.$$

The important point is that the algorithm always computes the unified Kaplan–Meier estimator. Probes are not a separate learning phase; they are uncensored observations added to the same KM sample. The bracketing and probe certificates guarantee that the confidence set remains valid even when the KM risk sets are weak.

4.3. Arm pseudo-cost estimates and confidence bounds

We now define the empirical objective used to compare candidate base-stock levels. Recall that after m completed episodes, the shrinkage confidence set is $\mathcal{C}_\mu(m)$, with midpoint and radius

$$\hat{\mu}_m := \mathrm{mid}(\mathcal{C}_\mu(m)), \qquad r_\mu(m) := \frac{1}{2}\,\mathrm{diam}(\mathcal{C}_\mu(m)).$$

On the confidence event proved in Section 5,

$$|\hat{\mu}_m - \mu_\rho| \le r_\mu(m) \qquad \text{for all } m.$$

Only completed decision cycles are used to estimate the performance of the candidate base-stock levels. For arm $i \in \{1, \dots, M\}$, define

$$\mathcal{I}_i(m) := \{\ell \le m : \text{episode } \ell \text{ is a completed decision cycle at } s_i\}, \qquad N_i(m) := |\mathcal{I}_i(m)|.$$

For each $\ell \in \mathcal{I}_i(m)$, the algorithm records

$$\Lambda_\ell, \qquad Q_\ell, \qquad V_\ell, \qquad W_\ell := Q_\ell - V_\ell,$$

where Λ_ℓ is the cycle length, Q_ℓ is the total order quantity, V_ℓ is total sales, and W_ℓ is the observable total physical shrinkage in the cycle.

For a completed adaptive episode ℓ, define its realized pseudo-cost evaluated at coefficient μ by

$$\widetilde{K}_\ell^{\mathrm{obs}}(\mu) := \alpha(\mu) W_\ell - (h+b) V_\ell. \tag{25}$$

If $\ell \in \mathcal{I}_i(m)$, then this realized cycle was generated by arm s_i and by the tracker parameter that was available at the start of that episode,

$$\nu_{\ell-1} := \hat{\mu}_{\ell-1}.$$

The empirical pseudo-cost rate of arm i is

$$\hat{\gamma}_i(m) := \frac{\sum_{\ell \in \mathcal{I}_i(m)} \widetilde{K}_\ell^{\mathrm{obs}}(\hat{\mu}_m)}{\sum_{\ell \in \mathcal{I}_i(m)} \Lambda_\ell}, \qquad N_i(m) > 0. \tag{26}$$

The denominator is total observed cycle time. This is important: the population objective is a regenerative reward rate, so cycles are weighted by their lengths rather than averaged as if they had equal duration.

We attach a confidence radius

$$B_i(m) := B_i^{\mathrm{stat}}(m) + B_i^{\mu,\mathrm{eval}}(m) + B_i^{\mu,\mathrm{impl}}(m). \tag{27}$$

The three terms correspond to three different errors: sampling error in the cycle pseudo-costs, plug-in error from evaluating pseudo-costs at $\hat{\mu}_m$, and implementation error from having executed past cycles with earlier tracker estimates.

Sampling term. Let

$$L_{M,T} := \log(64MT^5), \qquad \bar{\Lambda}_T := \left\lceil \frac{1}{p_0}\log(8T^5) \right\rceil.$$

Also define the per-period pseudo-cost envelope

$$\bar{c}_{\mathrm{per}} := \alpha(\bar{\mu})\bar{S} + \big(\alpha(\bar{\mu}) + h + b\big)\overline{D},$$

and

$$c_{\mathrm{stat},T} := 2\sqrt{2}\,\bar{c}_{\mathrm{per}}\bar{\Lambda}_T.$$

The statistical radius is

$$B_i^{\mathrm{stat}}(m) := c_{\mathrm{stat},T}\sqrt{\frac{L_{M,T}}{N_i(m)}}, \qquad N_i(m) > 0. \tag{28}$$

Section 5 proves that this term uniformly controls the martingale fluctuations of the observed cycle pseudo-cost rates for all arms and all episode counts up to the horizon.

Current evaluation error. The pseudo-cost coefficient is

$$\alpha(\mu) = \frac{h\mu}{1-\mu}.$$

On $[\underline{\mu}, \bar{\mu}]$, it is Lipschitz with constant

$$L_\alpha := \frac{h}{(1-\bar{\mu})^2}. \tag{29}$$

Therefore, on the shrinkage confidence event,

$$|\alpha(\hat{\mu}_m) - \alpha(\mu_\rho)| \le L_\alpha r_\mu(m).$$

Moreover, for every completed cycle,

$$0 \le W_\ell \le Q_\ell \le \bar{S}\Lambda_\ell.$$

Thus the plug-in error caused by evaluating all past pseudo-costs with $\alpha(\hat{\mu}_m)$ rather than $\alpha(\mu_\rho)$ is bounded by

$$\left| \frac{\sum_{\ell\in\mathcal{I}_i(m)}[\alpha(\hat{\mu}_m) - \alpha(\mu_\rho)]W_\ell}{\sum_{\ell\in\mathcal{I}_i(m)}\Lambda_\ell} \right| \le L_\alpha\bar{S}\,r_\mu(m).$$

We set

$$B_i^{\mu,\mathrm{eval}}(m) := L_\alpha\bar{S}\,r_\mu(m). \tag{30}$$

Past implementation error. Cycle ℓ was not necessarily executed with the correct tracker parameter μ_ρ. It was executed with the estimate available at the start of that episode,

$$\nu_{\ell-1} := \hat{\mu}_{\ell-1}.$$

On the confidence event,

$$|\nu_{\ell-1} - \mu_\rho| \le r_\mu(\ell - 1).$$

The implementation-stability lemma, Lemma 8, states that there is a finite primitive constant L_{impl} such that

$$|g(S;\nu) - g(S;\nu')| \le L_{\mathrm{impl}}|\nu - \nu'| \qquad \forall S \in [0, \bar{S}], \quad \nu, \nu' \in [\underline{\mu}, \bar{\mu}].$$

Hence the average implementation error in the cycles used to estimate arm i is controlled by the cycle-length-weighted average of the shrinkage radii that were available when those cycles were actually run:

$$B_i^{\mu,\mathrm{impl}}(m) := L_{\mathrm{impl}} \frac{\sum_{\ell \in \mathcal{I}_i(m)} \Lambda_\ell \, r_\mu(\ell - 1)}{\sum_{\ell \in \mathcal{I}_i(m)} \Lambda_\ell}, \qquad N_i(m) > 0. \tag{31}$$

If the notation $r_\mu(0)$ is needed before the first confidence interval is formed, set $r_\mu(0) := (\bar{\mu} - \underline{\mu})/2$. In the implemented algorithm, the mandatory warm-start manual count~~probe~~[MG] creates the first nontrivial confidence set before any decision cycle is run.

For an unsampled active arm, we set

$$\mathrm{LCB}_i(m) := -\infty, \qquad \mathrm{UCB}_i(m) := +\infty,$$

so that every active arm is sampled at least once before it can be eliminated. For $N_i(m) > 0$, define

$$\mathrm{LCB}_i(m) := \hat{\gamma}_i(m) - B_i(m), \qquad \mathrm{UCB}_i(m) := \hat{\gamma}_i(m) + B_i(m). \tag{32}$$

4.4. Adaptive probe schedule

The algorithm uses a shrinking uncertainty target rather than a fixed count~~probe~~[MG] frequency. A clean probe is an information-generating action: the retailer places a token replenishment order s_{pr} at the stockout boundary, operates normally for one period, and then conducts a complete manual count of the remaining on-shelf inventory at the period's end. The inventory balance $\rho_t s_{\mathrm{pr}} = Y_t + X_{t+1}$ then yields the exact shrinkage realization ρ_t regardless of whether a stockout occurred. Operationally, this corresponds to a dedicated counting shift—an inventory audit or reconciliation period—that the retailer initiates whenever the shrinkage uncertainty is high enough to materially distort replenishment decisions. The probe is useful because it provides an uncensored draw of ρ_t, but costly because the labor and time required for the physical count displace normal warehouse operations for one period. We model the cost of one manual count as C_{pr}.~~A clean probe is an information-generating action: the retailer conducts a complete manual count of the warehouse at the current stockout boundary, places the token order $s_{\mathrm{pr}} < \underline{D}$, and reads off the exact shrinkage realization ρ_t at the close of the period. Operationally, this corresponds to a dedicated counting shift—an inventory audit or reconciliation period—that the retailer initiates whenever the shrinkage uncertainty is high enough to materially distort replenishment decisions. The probe is useful because it provides an uncensored draw of ρ_t, but costly because the labor and time required for the physical count displace normal warehouse operations for one period. We model the cost of one manual count as C_{pr}.~~MG

The relevant quantity is the certified shrinkage radius $r_\mu(m)$. If the algorithm starts decision episodes while $r_\mu(m)$ is large, then both the pseudo-cost evaluation and the tracker implementation are inaccurate. The regret proof makes this precise: the shrinkage-learning contribution has the form

$$C_{\mathrm{pr}} P_T + C_{\mathrm{sh,T}} \sum_{m \in \mathcal{D}_T} r_m^-, \qquad r_m^- := r_\mu(m-1),$$

where C_{pr} is the cost of one manual count (modeled as an upper bound on the labor and time cost of a full physical inventory audit)~~$C_{\mathrm{pr}} := b\overline{D}$ bounds the one-period cost of a clean probe~~MG, and $C_{\mathrm{sh,T}}$ is the sensitivity constant multiplying cumulative shrinkage uncertainty in the upper bound.

This display explains the threshold. For a generic nonincreasing target sequence b_m, the manual-count~~probe~~MG certificate implies that maintaining radius b_T by time T requires on the order of

$$\frac{L_T}{b_T^2}$$

manual counts~~clean probes~~MG. Meanwhile, if decision episodes start with radius at most b_m, the cumulative shrinkage-inaccuracy term is controlled by

$$\sum_{m=1}^{T} b_m.$$

Thus the proof-level shrinkage-learning contribution is governed by

$$C_{\mathrm{pr}} \frac{L_T}{b_T^2} + C_{\mathrm{sh,T}} \sum_{m=1}^{T} b_m.$$

Taking b_m proportional to $(L_T/m)^{1/3}$ makes the two terms have the same order:

$$\frac{L_T}{b_T^2} \asymp L_T^{1/3} T^{2/3}, \qquad \sum_{m=1}^{T} b_m \asymp L_T^{1/3} T^{2/3}.$$

For the horizon-dependent analysis, set

$$L_T := \log(384T^6),$$

and use the normalized robust schedule

$$a_m := \min\left\{1, \left(\frac{L_T}{m}\right)^{1/3}\right\}, \qquad m \geq 1. \tag{33}$$

The clipping at one says that, at very early times, the algorithm does not try to force the shrinkage interval below the trivial unit scale.

At the beginning of episode $m+1$, after computing the combined shrinkage confidence interval, the algorithm applies the rule

$$\text{probe if and only if} \qquad r_\mu(m) > a_{m+1}. \tag{34}$$

Otherwise it starts a decision cycle.

The rule is overlap-adaptive. If ordinary episode-start observations already provide enough shrinkage information through the Kaplan–Meier certificate, then $r_\mu(m)$ may fall below the critical radius without any manual count~~clean probe~~[MG], and the algorithm counts~~audits~~[MG] less. If natural overlap is weak, the manual-count~~probe~~[MG] certificate provides the robust fallback. In the worst case, the threshold keeps both the number of manual counts~~number of probes~~[MG] and the cumulative shrinkage radius at the same $\widetilde{O}(T^{2/3})$ scale.

4.5. Proposed Algorithm

The algorithm starts with one manual count (clean probe). This warm start is not conceptually necessary, but it ensures that the first tracker update is anchored by at least one complete, uncensored shrinkage observation before any replenishment cycle is run, and costs only the labor of one physical count. Thereafter the algorithm alternates adaptively between decision cycles and additional manual counts.~~The algorithm starts with one clean probe. This warm start is not conceptually necessary, but it ensures that the first tracker update is anchored by at least one complete shrinkage observation and costs only one period. Thereafter the algorithm alternates adaptively between decision cycles and additional probes.~~[MG] Arm selection is optimistic: among non-eliminated arms,

Algorithm 1: KM-Probe optimistic learning for hidden shrinkage

Input: Grid $\mathcal{S}_M = \{s_1, \ldots, s_M\} \subset (0, \bar{S}]$, token order level for manual-count (probe) episodes $s_{\mathrm{pr}} > 0$~~token order level for manual-count (probe) episodes $s_{\mathrm{pr}} \le \underline{D}$~~$^{\mathrm{MG}}$, horizon T, confidence schedule $(\delta_m)_{m \ge 0}$, target schedule $(a_m)_{m \ge 1}$, and finite KM truncation grid $\mathcal{A}_T \subset [0, 1]$.

Output: Recommended base-stock level $\hat{S}_T$.

Initialize active set $\mathcal{A} \leftarrow \{1, \ldots, M\}$, shrinkage log $\mathscr{D}_\rho(0) \leftarrow \emptyset$, and arm logs $\mathcal{I}_i(0) \leftarrow \emptyset$ for all i.

Set completed-episode counter $m \leftarrow 0$ and calendar time $t \leftarrow 1$.

Conduct one manual count (clean probe): place token order s_{pr}; at period's end observe sales Y_t, count ending inventory X_{t+1}, recover $\rho_t = (Y_t + X_{t+1})/s_{\mathrm{pr}}$, and append $(\rho_t, 1)$ to $\mathscr{D}_\rho$~~Conduct one manual count (clean probe): place token order s_{pr}, observe (Y_t, Z_t) at period's end, append $(Y_t/s_{\mathrm{pr}}, Z_t)$ to $\mathscr{D}_\rho$~~$^{\mathrm{MG}}$, set $t \leftarrow t + 1$, and set $m \leftarrow 1$.

while $t \le T$ **do**

  Compute the shrinkage confidence set $\mathcal{C}_\mu(m)$ from $\mathscr{D}_\rho(m)$ using (23).

  Set $\hat{\mu}_m \leftarrow \mathrm{mid}(\mathcal{C}_\mu(m))$ and $r_\mu(m) \leftarrow \mathrm{diam}(\mathcal{C}_\mu(m))/2$.

  For each active arm $i \in \mathcal{A}$ with $N_i(m) > 0$, compute $\hat{\gamma}_i(m)$ from (26), $B_i(m)$ from (27), and $(\mathrm{LCB}_i(m), \mathrm{UCB}_i(m))$ from (32).

  Eliminate every active arm i such that

$$\mathrm{LCB}_i(m) > \min_{j \in \mathcal{A}} \mathrm{UCB}_j(m).$$

  Update $\mathcal{A}$ to the surviving active set.

  if $r_\mu(m) > a_{m+1}$ **then**

    Conduct one manual count (clean probe): place token order s_{pr}; at period's end observe sales Y_t, count ending inventory X_{t+1}, recover $\rho_t = (Y_t + X_{t+1})/s_{\mathrm{pr}}$, and append $(\rho_t, 1)$ to $\mathscr{D}_\rho$~~Conduct one manual count (clean probe): place token order s_{pr}, observe (Y_t, Z_t) at period's end, append $(Y_t/s_{\mathrm{pr}}, Z_t)$ to $\mathscr{D}_\rho$~~$^{\mathrm{MG}}$.

  else

    if *there exists* $i \in \mathcal{A}$ *with* $N_i(m) = 0$ **then**

      Select any unsampled active arm $i_m \in \mathcal{A}$ with $N_{i_m}(m) = 0$.

    else

      Select an optimistic arm

$$i_m \in \arg\min_{i \in \mathcal{A}} \mathrm{LCB}_i(m).$$

    end

    Start a decision episode at level s_{i_m} using tracker parameter $\nu_m = \hat{\mu}_m$.

    At the first period, append $(Y_t/s_{i_m}, Z_t)$ to $\mathscr{D}_\rho$.

    Continue the episode until the next stockout and record the completed-cycle quantities $(\Lambda_m, Q_m, V_m, W_m)$; append m to $\mathcal{I}_{i_m}$.

  end

  Advance t to the first period after the completed episode and set $m \leftarrow m + 1$.

end

Return

$$\hat{S}_T \in \arg\min_{i \in \mathcal{A}:\, N_i(m) > 0} \hat{\gamma}_i(m),$$

with ties broken arbitrarily.

it plays the arm with the smallest lower confidence bound. Elimination is used only as safe pruning: an arm is removed once its lower confidence bound is above the best upper confidence bound among active competitors.

The algorithm is easiest to read by following the two logs separately. The shrinkage log $\mathscr{D}_\rho$ grows by one observation at every episode start, whether that episode is a manual count~~probe~~MG or a genuine decision cycle. This log is the only object used to learn μ_ρ. The arm logs $\mathcal{I}_i$ grow only when the algorithm actually plays arm i through a full decision cycle; these logs determine the pseudo-cost estimates and confidence intervals used for arm comparison.

The manual-count rule~~The probe rule~~MG in (34) is equally important. If natural overlap is strong, the combined confidence interval for μ_ρ shrinks rapidly and the algorithm soon spends almost all of its effort comparing arms. If censoring is severe, the confidence radius remains too wide, the threshold test fails, and the algorithm pays for complete shrinkage observations through manual counts~~pays for complete shrinkage observations through clean probes~~MG. In this sense the method is not "KM or manual counts~~probes~~MG." It is a single learning procedure in which manual counts are simply high-quality, uncensored additions~~probes are simply high-quality additions~~MG to the same censored sample.

Finally, lower-confidence-bound selection is paired with safe elimination for a reason. Pure round-robin elimination is statistically wasteful once the active set narrows, while aggressive optimism without elimination keeps clearly inferior arms alive longer than necessary. The present combination preserves the clean confidence-based logic needed for analysis while still directing most data to the most promising arms.

5. Regret Analysis

The regret analysis separates three sources of error. First, the algorithm only searches over the finite grid $\mathcal{S}_M$, even though the benchmark is the continuous oracle $S^\star_{\mathrm{EV}}$. Second, the grid arms have to be learned from regenerative pseudo-cost observations. Third, the nuisance parameter μ_ρ affects both pseudo-cost evaluation and the tracker used to execute a policy. The theorem below includes the deterministic discretization term directly, so the stated bound is already a regret bound against the continuous benchmark in (5).

5.1. Main upper bound

The upper bound is derived under the following primitive regularity conditions.

Theorem 1 (Robust upper bound) *Under Assumptions 1–4, run the algorithm of Section 4 on the uniform grid*

$$\mathcal{S}_M := \left\{ \frac{j\bar{S}}{M} : j = 1, \ldots, M \right\}.$$

We have

$$R_T = \widetilde{O}(T^{2/3}),$$

where the hidden logarithmic factors depend only on the model primitives.

The theorem is a worst-case guarantee. The proof first establishes a sharper grid-level inequality in terms of the number of probes and the realized shrinkage-radius path $(r_\mu(m))$. That intermediate inequality is useful for interpretation: when ordinary episode-start observations make the Kaplan–Meier certificate shrink quickly, the algorithm probes less and the nuisance-parameter terms are smaller on a fixed grid. The robust bound above does not require any such overlap condition. After the grid is optimized against the continuous benchmark, the worst-case rate is $\widetilde{O}(T^{2/3})$, matching the primitive lower bound in Section 5.2.

Interpretation of the shrinkage-learning term. The robust $T^{2/3}$ term has a concrete audit interpretation. The algorithm faces two opposing costs. A clean probe is an audit: it gives an uncensored shrinkage observation, but it costs a selling period. Avoiding audits is also costly, because the expected-inventory tracker and the pseudo-cost coefficient both depend on μ_ρ. If the shrinkage radius is r, the resulting evaluation and implementation errors are linear in r. Thus the regret analysis contains the two quantities

$$C_{\mathrm{pr}} P_T \qquad \text{and} \qquad C_{\mathrm{sh}} \sum_{m \in \mathcal{D}_T} r_m^-,$$

where P_T is the number of probes, C_{pr} is the direct probe cost and C_{sh} is the constant multiplying the shrinkage-radius terms in the master bound.

The threshold schedule

$$a_m \asymp (L_T/m)^{1/3}$$

is the critical curve that keeps these two terms at the same order. If the radius is above the curve, the algorithm audits. If the radius is below the curve, the algorithm operates. Natural stockout observations enter through the Kaplan–Meier certificate and may lower $r_\mu(m)$ without any audit; in that case the algorithm automatically audits less. The clean probes are therefore a robust fallback, not a separate exploration phase.

5.2. Lower Bound: A Primitive Audited-Reset Hard Instance

We now prove a primitive minimax lower bound on a granular audited-reset subclass of the hidden-shrinkage model. The subclass is deliberately simple: demand is deterministic, the admissible base-stock levels are granular, and the system is audited and reset to zero after each period. Thus every period is an episode start. This makes the learner's state information as favorable as possible: the learner never has to reason about hidden inventory carried across periods. The lower bound therefore isolates the information cost of learning hidden shrinkage.

The audited-reset feature is used only for the lower bound. It removes continuation-period leakage and makes the hard instance fully transparent. The difficulty remains: ordinary operating levels do not reveal the hard shrinkage bit, while the separated probe level reveals it but is costly. This is the hidden-shrinkage analogue of active-exploration lower bounds in granular censored-demand learning, where the learner must deliberately choose an information-generating action to distinguish statistically close instances.

5.3. Audited-reset hard subclass

In the hard subclass, every period starts from

$$X_t = 0, \qquad \hat{X}_t = 0.$$

The learner chooses one of three granular base-stock levels

$$\mathcal{S}_{\text{hard}} := \{S_P, S_L, S_H\}.$$

At the end of the period the system is audited and reset to zero. Thus every period is a one-period episode. The learner observes only sales and the stockout indicator, not the realized shrinkage, lost demand, true inventory, or cost.

We fix the following primitive constants:

$$d = 1, \qquad h = 1, \qquad b = 3,$$

and the shrinkage support

$$\rho_t \in \left\{0, \frac{1}{4}, \frac{1}{2}, 1\right\}.$$

The demand distribution is deterministic:

$$D_t \equiv 1.$$

The three admissible base-stock levels are

$$S_P := \frac{1}{2}, \qquad S_L := 2, \qquad S_H := 4.$$

The level S_P is the separated probe level, S_L is the low operating level, and S_H is the high operating level.

For a small parameter $\Delta > 0$, define two shrinkage laws P_Δ^+ and P_Δ^-. The common masses are

$$\mathbb{P}_\pm(\rho = 0) = \frac{1}{5}, \qquad \mathbb{P}_\pm\left(\rho = \frac{1}{4}\right) = \frac{2}{5}.$$

The hard bit is carried by the split between $\rho = 1/2$ and $\rho = 1$:

$$\mathbb{P}_+\left(\rho = \frac{1}{2}\right) = \frac{1}{5} - 2\Delta, \qquad \mathbb{P}_+(\rho = 1) = \frac{1}{5} + 2\Delta,$$

and

$$\mathbb{P}_-\left(\rho = \frac{1}{2}\right) = \frac{1}{5} + 2\Delta, \qquad \mathbb{P}_-(\rho = 1) = \frac{1}{5} - 2\Delta.$$

For $\Delta \leq 1/10$, these are valid probability laws. The corresponding mean survival factors satisfy

$$\mu_\rho(P_\Delta^+) = \frac{2}{5} + \Delta, \qquad \mu_\rho(P_\Delta^-) = \frac{2}{5} - \Delta.$$

The one-period cost at base-stock level S is

$$c(S, \rho) := h(\rho S - 1)^+ + b(1 - \rho S)^+.$$

The observation is

$$O(S, \rho) := \left(\min\{\rho S, 1\},\ \mathbf{1}\{1 > \rho S\}\right).$$

The three levels have the following roles.

Probe level. At $S_P = 1/2$, every positive shrinkage state stocks out:

$$\rho S_P < 1 \qquad \text{for } \rho \in \left\{\frac{1}{4}, \frac{1}{2}, 1\right\}.$$

Therefore

$$Y = \rho S_P, \qquad Z = 1,$$

and the learner observes $\rho = Y/S_P$ exactly. Hence the probe level distinguishes P_Δ^+ from P_Δ^-.

Operating levels. At $S_L = 2$,

$$\rho = \frac{1}{2} \quad \text{and} \quad \rho = 1$$

both produce

$$(Y, Z) = (1, 0).$$

At $S_H = 4$,

$$\rho = \frac{1}{4}, \frac{1}{2}, 1$$

all produce

$$(Y, Z) = (1, 0).$$

Thus the hard split between $1/2$ and 1 is invisible at both operating levels. At S_L, the observation law distinguishes only the common states $\rho = 0$, $\rho = 1/4$, and the aggregate event $\rho \in \{1/2, 1\}$. At S_H, it distinguishes only $\rho = 0$ from $\rho > 0$. Both observation laws are identical under P_Δ^+ and P_Δ^-.

Operating cost switch. The operating levels S_L and S_H have opposite optimality under the two instances. A direct calculation gives

$$c(S_L, 0) = 3, \quad c\left(S_L, \frac{1}{4}\right) = \frac{3}{2}, \quad c\left(S_L, \frac{1}{2}\right) = 0, \quad c(S_L, 1) = 1,$$

and

$$c(S_H, 0) = 3, \quad c\left(S_H, \frac{1}{4}\right) = 0, \quad c\left(S_H, \frac{1}{2}\right) = 1, \quad c(S_H, 1) = 3.$$

Consequently,

$$\mathbb{E}_+[c(S_H, \rho) - c(S_L, \rho)] = 2\Delta,$$

while

$$\mathbb{E}_-[c(S_L, \rho) - c(S_H, \rho)] = 2\Delta.$$

Thus S_L is the unique optimal operating level under P_Δ^+, and S_H is the unique optimal operating level under P_Δ^-, with operating gap 2Δ.

The probe level is separated and costly. For all sufficiently small $\Delta \le 1/10$,

$$\mathbb{E}_\sigma[c(S_P, \rho)] - \min\{\mathbb{E}_\sigma[c(S_L, \rho)], \mathbb{E}_\sigma[c(S_H, \rho)]\} \ge c_P \qquad \text{for } \sigma \in \{+, -\},$$

for a numerical constant $c_P > 0$.

Let Π_{ar} denote the set of nonanticipative policies for this audited-reset subproblem, i.e., policies that choose an action $J_t \in \{P, L, H\}$ based only on the past observations. For $P \in \{P_\Delta^+, P_\Delta^-\}$, regret is measured against the best fixed action in $\mathcal{S}_{\mathrm{hard}}$:

$$R_T^\pi(P) := \mathbb{E}_P^\pi\left[\sum_{t=1}^{T} c(S_{J_t}, \rho_t)\right] - T \min_{j \in \{P, L, H\}} \mathbb{E}_P[c(S_j, \rho)].$$

Theorem 2 (Primitive audited-reset lower bound) *For the audited-reset hard subclass above, there exist constants $c > 0$, $a > 0$, and $T_0 < \infty$ such that for all $T \geq T_0$, with*

$$\Delta_T := aT^{-1/3},$$

we have

$$\inf_{\pi \in \Pi_{\mathrm{ar}}} \sup_{\sigma \in \{+,-\}} R_T^\pi(P_{\Delta_T}^\sigma) \geq cT^{2/3}.$$

Consequently, any lower-bound model class containing this primitive audited-reset granular subproblem has minimax regret at least $\Omega(T^{2/3})$.

The proof is a two-point testing argument. Operating levels generate zero information about the hard shrinkage split. The probe level generates $O(\Delta^2)$ KL information per use and has a constant regret price. If a policy probes often enough to identify the sign of $\mu_\rho - 2/5$, it pays order Δ^{-2}. If it does not probe often enough, the two instances remain statistically indistinguishable and the policy chooses the wrong operating level for a constant fraction of periods, paying order $T\Delta$. Balancing

$$\Delta^{-2} \qquad \text{and} \qquad T\Delta$$

gives $\Delta \asymp T^{-1/3}$ and regret $\Omega(T^{2/3})$.

6. Numerical Analysis

7. Conclusion

Motivated by a collaboration with a small-scale online grocer operating a large perishable assortment under severe traceability constraints, this paper studies periodic-review inventory management when the true on-hand stock is unobservable because hidden stochastic shrinkage—spoilage, theft, physical damage, and record error—continuously erodes usable inventory between replenishment cycles. The retailer observes only realized sales and stockout indicators; the true inventory level, lost demand, and the shrinkage realization in each period remain hidden.

We formulate the problem as a periodic-review system in which usable inventory is determined by an unknown random survival factor ρ_t applied to on-hand stock before demand is realized. We develop an implementable **KM-Probe** algorithm which operates over a family of expected-inventory base-stock policies. At every stockout event, the algorithm faces a choice: launch a decision cycle to gather pseudo-cost evidence for a candidate base-stock level, or conduct a clean probe—a period in which a token replenishment order is placed and a complete manual count of the ending inventory is conducted, recovering the exact shrinkage draw. We establish a $\widetilde{O}(T^{2/3})$

regret upper bound against the best continuous base-stock level $S^{\star}_{\mathrm{EV}}$, with the $T^{2/3}$ rate arising from balancing the labor cost of conducting manual counts against the shrinkage-inaccuracy cost of running the tracker with an imprecise μ_ρ estimate.

Implementability for small-scale retailers. A central design objective throughout this paper is that every statistic and decision required by the algorithm can be computed with the simple data infrastructure that small-scale retailers actually possess. Namely, KM-probe algorithm involves only running sums of order quantities and sales, which are already tracked in even the most basic spreadsheet system. Furthermore, the adaptive probe decision reduces to a single comparison: is the current certified shrinkage confidence interval $r_\mu(m)$ above a pre-computed threshold? In this way, the KM-Probe algorithm replicates, at the level of statistical precision required for near-optimal replenishment, the inventory traceability that large retailers achieve through expensive ERP and RFID infrastructure—using nothing more than order records, sales totals, stockout flags, and the occasional physical count.

Managerial implications. Our results yield several managerial implications. First of all, the adaptive probe schedule implies that the value of the algorithm and the optimal frequency of manual counts depend directly on the shrinkage characteristics of the product category being managed. The mean survival factor μ_ρ and its variability—which together determine how quickly natural stockout observations sharpen the Kaplan–Meier shrinkage estimate—differ substantially across the categories relevant to our partner retailer, and the practical recommendations differ accordingly.

The data collected from our industrial partner shows that *fresh vegetables and salad greens* exhibit rapid, highly variable spoilage driven by temperature fluctuations, supplier heterogeneity, and short shelf lives of two to five days. The survival factor ρ_t is therefore highly variable and its mean changes across supplier batches and seasons, making shrinkage uncertainty persistently high. The KM-Probe algorithm will trigger frequent manual counts for this category, as natural stockout observations provide only weakly censored draws and the Kaplan–Meier certificate shrinks slowly. On the other hand, *dairy products* such as milk, yogurt, and cheese exhibit more predictable spoilage tied to printed expiry dates, but are still subject to cold-chain failures and handling losses that introduce variability in ρ_t. Stockouts are relatively frequent in this category because demand is high and orders are often tight, which means the Kaplan–Meier certificate benefits from a denser stream of uncensored observations. Last but note least, *packaged beverages* present the most favorable environment for the algorithm. Shrinkage in this category is primarily driven by breakage rather than spoilage, and the survival factor is typically high and stable—μ_ρ close to one with low variance. When μ_ρ is large and stable, natural stockout observations provide tightly censored but still informative draws of ρ_t, and the Kaplan–Meier certificate tends to converge quickly. The algorithm will probe infrequently for beverages, and the regret in this category will be dominated by the bandit-learning term rather than the shrinkage-learning term.

Extensions. Several directions for future work remain open. An important next step is to incorporate product-specific covariates—storage temperature, supplier identity, days since delivery—into the shrinkage model, allowing the survival factor distribution to depend on observable features and potentially enabling faster learning. A second direction is to extend the analysis to settings with positive replenishment lead times, where orders placed in one period affect usable inventory several periods later and the regenerative structure of stockout cycles becomes more complex. Finally, a multi-item extension that coordinates the probe schedule across SKUs—exploiting the fact that counting one product in a shelf section is often bundled with counting adjacent products—could reduce the effective cost C_{pr} and shift the optimal probe frequency across the entire assortment simultaneously.

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Appendix A: Proof of Proposition 1

Fix $S \in [0, \bar{S}]$ and $\nu \in (0,1)$, and suppress the dependence on (S,ν) in the period-level notation. All expectations are taken under the true dynamics induced by the fixed policy π_S^ν. We use the standard regenerative regularity condition that the stockout cycles have finite mean length and finite first moments of the cycle rewards. In particular,

$$0 < \mathbb{E}[\Lambda_1^{S,\nu}] < \infty, \qquad \mathbb{E}[V_1^{S,\nu} + W_1^{S,\nu}] < \infty,$$

so that the renewal-reward theorem applies.

Let

$$B_t := X_t + q_t$$

denote the pre-shrinkage inventory, and define the physical shrinkage in period t by

$$R_t^{\mathrm{sh}} := (1-\rho_t)B_t.$$

Then

$$A_t = \rho_t B_t, \qquad X_{t+1} = B_t - R_t^{\mathrm{sh}} - Y_t.$$

Since $X_{t+1} = A_t - Y_t$ and $L_t = D_t - Y_t$, the one-period cost can be written as

$$C_t = hX_{t+1} + bL_t = h(A_t - Y_t) + b(D_t - Y_t) = bD_t + hA_t - (h+b)Y_t. \tag{1}$$

We next relate the expected usable inventory A_t to the expected physical shrinkage R_t^{sh}. Let $\mathcal{F}_{t-1}$ denote the history available just before period t. The quantity $B_t = X_t + q_t$ is $\mathcal{F}_{t-1}$-measurable, whereas ρ_t is independent of $\mathcal{F}_{t-1}$ and has mean μ_ρ. Moreover, for a cycle started from a stockout state, the event $\{t \le \Lambda_1^{S,\nu}\}$ depends only on periods before t, and is therefore $\mathcal{F}_{t-1}$-measurable. Hence

$$\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}A_t\right] = \mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}\rho_t B_t\right] = \mu_\rho\,\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}B_t\right],$$

while

$$\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}R_t^{\mathrm{sh}}\right] = \mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}(1-\rho_t)B_t\right] = (1-\mu_\rho)\,\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}B_t\right].$$

Since $\mu_\rho \in (0,1)$, summing over $t \ge 1$ gives

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} A_t\right] = \frac{\mu_\rho}{1-\mu_\rho}\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} R_t^{\mathrm{sh}}\right]. \tag{2}$$

Now use the inventory balance over a completed cycle. Because stockout periods reset the true inventory to zero, a completed cycle starts and ends with zero true inventory:

$$X_{\tau_0+1} = 0, \qquad X_{\tau_1+1} = 0.$$

Summing

$$X_{t+1} = X_t + q_t - R_t^{\mathrm{sh}} - Y_t$$

over $t \in \mathcal{C}_1$ yields

$$0 = Q_1^{S,\nu} - \sum_{t\in\mathcal{C}_1} R_t^{\mathrm{sh}} - V_1^{S,\nu}.$$

Therefore

$$\sum_{t\in\mathcal{C}_1} R_t^{\mathrm{sh}} = Q_1^{S,\nu} - V_1^{S,\nu} = W_1^{S,\nu}. \tag{3}$$

Combining (2) and (3),

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} A_t\right] = \frac{\mu_\rho}{1-\mu_\rho}\,\mathbb{E}[W_1^{S,\nu}]. \tag{4}$$

Next, by Wald's identity applied to the i.i.d. demand sequence,

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} D_t\right] = \mathbb{E}[D_t]\,\mathbb{E}[\Lambda_1^{S,\nu}]. \tag{5}$$

Using (1), (4), and (5), the expected total true cost in one completed cycle is

$$\begin{aligned}\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} C_t\right] &= b\,\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} D_t\right] + h\,\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} A_t\right] - (h+b)\,\mathbb{E}[V_1^{S,\nu}] \\ &= b\,\mathbb{E}[D_t]\,\mathbb{E}[\Lambda_1^{S,\nu}] + \frac{h\mu_\rho}{1-\mu_\rho}\,\mathbb{E}[W_1^{S,\nu}] - (h+b)\,\mathbb{E}[V_1^{S,\nu}].\end{aligned}$$

Since

$$\alpha(\mu_\rho) := \frac{h\mu_\rho}{1-\mu_\rho},$$

this becomes

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} C_t\right] = b\,\mathbb{E}[D_t]\,\mathbb{E}[\Lambda_1^{S,\nu}] + \alpha(\mu_\rho)\,\mathbb{E}[W_1^{S,\nu}] - (h+b)\,\mathbb{E}[V_1^{S,\nu}]. \tag{6}$$

By the renewal-reward theorem,

$$g(S;\nu) = \frac{\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} C_t\right]}{\mathbb{E}[\Lambda_1^{S,\nu}]}.$$

Dividing (6) by $\mathbb{E}[\Lambda_1^{S,\nu}]$ gives

$$g(S;\nu) = b\,\mathbb{E}[D_t] + \alpha(\mu_\rho)\,\frac{\mathbb{E}[W_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]} - (h+b)\,\frac{\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}. \tag{7}$$

Because the stockout cycles are regenerative and identically distributed under the fixed policy π_S^ν,

$$w(S;\nu) = \frac{\mathbb{E}[W_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}, \qquad v(S;\nu) = \frac{\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}.$$

Substituting these identities into (7) proves

$$g(S;\nu) = b\,\mathbb{E}[D_t] + \alpha(\mu_\rho)\,w(S;\nu) - (h+b)\,v(S;\nu).$$

It remains to verify the equivalent pseudo-cost representation. Under the fixed policy π_S^ν, the cycle pseudo-cost random variable is

$$\widetilde{K}_k^{S,\nu}(\mu_\rho) = \alpha(\mu_\rho)W_k^{S,\nu} - (h+b)V_k^{S,\nu}.$$

Since the completed cycles are identically distributed under the fixed policy,

$$\begin{aligned}\gamma(S;\nu) &= \frac{\mathbb{E}[\widetilde{K}_1^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda_1^{S,\nu}]} \\ &= \frac{\alpha(\mu_\rho)\mathbb{E}[W_1^{S,\nu}] - (h+b)\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]} \\ &= \alpha(\mu_\rho)w(S;\nu) - (h+b)v(S;\nu).\end{aligned}$$

Therefore

$$g(S;\nu) = b\,\mathbb{E}[D_t] + \gamma(S;\nu).$$

For the benchmark tracker $\nu = \mu_\rho$, the additive term $b\,\mathbb{E}[D_t]$ is independent of S. Hence minimizing $g(S;\mu_\rho)$ over $S \in [0,\bar{S}]$ is equivalent to minimizing

$$\gamma(S;\mu_\rho)$$

over $S \in [0,\bar{S}]$. This proves the proposition.

Appendix B: Proofs for the regret analysis

B.1. Auxiliary episode bounds

We first record a few deterministic envelopes used repeatedly below.

Lemma 1 (Episode envelopes) *Under Assumption 1, for every period t and every admissible policy,*

$$q_t \le \bar{S}, \qquad X_t \le (t-1)\bar{S}, \qquad C_t \le ht\bar{S} + b\overline{D}.$$

Consequently,

$$\sum_{t=1}^{T} C_t \le \frac{h\bar{S}}{2} T(T+1) + b\overline{D}T.$$

Moreover, for every decision episode started from a stockout state,

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}} C_t\right] \le \frac{h\bar{S}}{p_0^2} + \frac{b\overline{D}}{p_0}.$$

B ecause $\hat{X}_t \ge 0$, the order rule (2) gives $q_t \le \bar{S}$. Since $\rho_t \le 1$ and $D_t \ge 0$,

$$X_{t+1} = (\rho_t(X_t + q_t) - D_t)^+ \le X_t + q_t \le X_t + \bar{S}.$$

Starting from $X_1 = 0$, induction yields $X_t \le (t-1)\bar{S}$. Therefore

$$C_t = hX_{t+1} + b(D_t - Y_t) \le ht\bar{S} + b\overline{D}.$$

Summing over $t = 1, \ldots, T$ gives the horizon envelope.

For the episode bound, let Λ denote the cycle length. By Lemma 2 below, $\mathbb{P}(\Lambda \ge t) \le (1-p_0)^{t-1}$. Hence

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}} C_t\right] \le \sum_{t\ge 1}(ht\bar{S} + b\overline{D})\mathbb{P}(\Lambda \ge t) \le \sum_{t\ge 1}(ht\bar{S} + b\overline{D})(1-p_0)^{t-1},$$

which evaluates to the displayed expression.

Lemma 2 (Geometric control of episode lengths) *Under Assumptions 1 and 3, every decision episode started from a stockout state has length stochastically dominated by a geometric random variable with success probability p_0. In particular,*

$$\mathbb{E}[\Lambda_m \mid \mathcal{G}_{m-1}] \le \frac{1}{p_0}$$

for every decision episode m, where $\mathcal{G}_{m-1}$ is the history up to the end of episode $m-1$. Moreover, with probability at least $1 - T^{-4}$, every episode initiated before horizon T has length at most $\bar{\Lambda}_T$.

[Proof of Lemma 2] If $\rho_t = 0$, then the usable inventory in period t is zero regardless of the current state or order size. Since Assumption 1 gives $D_t \ge \underline{D} > 0$, such a period necessarily stocks out. Therefore every period in a decision episode has conditional probability at least p_0 of ending the episode. This proves the geometric domination and the conditional mean bound $\mathbb{E}[\Lambda_m \mid \mathcal{G}_{m-1}] \le 1/p_0$.

At most T episodes can be initiated before horizon T, because every episode lasts at least one period. Thus

$$\mathbb{P}\Big(\exists \text{ episode initiated before } T \text{ with length } > \bar{\Lambda}_T\Big) \le T(1-p_0)^{\bar{\Lambda}_T} \le T\exp(-p_0\bar{\Lambda}_T) \le T^{-4},$$

by the definition of $\bar{\Lambda}_T$.

B.2. Confidence events for shrinkage learning

Lemma 3 (Bracketing certificate validity) *For every* $1 \le m \le T$,

$$\mathbb{P}\big(\mu_\rho \notin \mathcal{C}_{\rm br}(m)\big) \le 2\delta_m.$$

Consequently, under the schedule

$$\delta_m := \frac{1}{64T^4(m+1)^2},$$

we have

$$\mathbb{P}\big(\forall 1 \le m \le T:\ \mu_\rho \in \mathcal{C}_{\rm br}(m)\big) \ge 1 - T^{-4}.$$

F or every episode ℓ,

$$L_\ell = U_\ell \le \rho_{t_\ell} \le H_\ell \qquad \text{a.s.}$$

The episode-start order ξ_ℓ is predictable from the history before episode ℓ, while ρ_{t_ℓ} is fresh and independent of that history. Hence, writing $\mathcal{G}_{\ell-1}$ for the history up to the end of episode $\ell - 1$,

$$\mathbb{E}[\rho_{t_\ell} \mid \mathcal{G}_{\ell-1}] = \mu_\rho.$$

Therefore

$$\mathbb{E}[L_\ell - \mu_\rho \mid \mathcal{G}_{\ell-1}] \le 0, \qquad \mathbb{E}[\mu_\rho - H_\ell \mid \mathcal{G}_{\ell-1}] \le 0.$$

Thus

$$S_m^L := \sum_{\ell=1}^{m} (L_\ell - \mu_\rho)$$

and

$$S_m^H := \sum_{\ell=1}^{m} (\mu_\rho - H_\ell)$$

are supermartingales.

Moreover, their increments have range length at most one:

$$L_\ell - \mu_\rho \in [-\mu_\rho, 1 - \mu_\rho], \qquad \mu_\rho - H_\ell \in [\mu_\rho - 1, \mu_\rho].$$

The Hoeffding–Azuma inequality for supermartingales with conditionally bounded increments therefore gives, for any $r > 0$,

$$\mathbb{P}\big(\bar{L}_m - \mu_\rho > r\big) = \mathbb{P}(S_m^L > mr) \le \exp(-2mr^2),$$

and

$$\mathbb{P}\big(\mu_\rho - \bar{H}_m > r\big) = \mathbb{P}(S_m^H > mr) \le \exp(-2mr^2).$$

With

$$r_{\rm br}(m) := \sqrt{\frac{\log(6/\delta_m)}{2m}},$$

we obtain

$$\mathbb{P}\big(\bar{L}_m - \mu_\rho > r_{\rm br}(m)\big) \le \frac{\delta_m}{6},$$

and

$$\mathbb{P}\big(\mu_\rho - \bar{H}_m > r_{\rm br}(m)\big) \le \frac{\delta_m}{6}.$$

Hence

$$\mathbb{P}\big(\mu_\rho \notin \mathcal{C}_{\rm br}(m)\big) \le \frac{\delta_m}{3} \le 2\delta_m,$$

which proves the first claim.

For the simultaneous statement, use the union bound:

$$\mathbb{P}\left(\exists 1 \le m \le T : \mu_\rho \notin \mathcal{C}_{\rm br}(m)\right) \le \sum_{m=1}^{T} 2\delta_m.$$

Under the stated schedule,

$$\sum_{m=1}^{T} 2\delta_m \le \frac{1}{32T^4} \sum_{m=1}^{\infty} \frac{1}{(m+1)^2} \le T^{-4}.$$

Therefore

$$\mathbb{P}\big(\forall 1 \le m \le T :\ \mu_\rho \in \mathcal{C}_{\rm br}(m)\big) \ge 1 - T^{-4}.$$

Lemma 4 (Probe certificate validity) *For every $m \le T$,*

$$\mathbb{P}\big(\mu_\rho \notin \mathcal{C}_{\rm pr}(m)\big) \le 2\delta_m.$$

Consequently,

$$\mathbb{P}\big(\forall m \le T :\ \mu_\rho \in \mathcal{C}_{\rm pr}(m)\big) \ge 1 - T^{-4}.$$

E numerate the probe observations in the order in which they occur:

$$\widehat{\rho}_1^{\rm pr}, \widehat{\rho}_2^{\rm pr}, \ldots.$$

Each probe is launched from a stockout state, and by Assumption 1 its first period stocks out immediately and reveals $\widehat{\rho}_j^{\rm pr} = \rho_{t_j}$. Since future shrinkage draws are independent of the past, the probe sample is i.i.d. in $[0,1]$ with mean μ_ρ. Hoeffding's inequality therefore gives the pointwise bound, and a union bound over the at most T possible probe counts yields the simultaneous claim.

Lemma 5 (Product-integral perturbation) *Let H and $\widehat{H}$ be cumulative hazard measures on $[0,a]$, with jumps in $[0,1]$. Let*

$$S(x) := \prod_{u \le x} (1 - dH(u)), \qquad \widehat{S}(x) := \prod_{u \le x} (1 - d\widehat{H}(u))$$

denote the associated product-integral survival functions. If

$$\sup_{0 \le x \le a} |\widehat{H}(x) - H(x)| \le \zeta,$$

then

$$\sup_{0 \le x \le a} |\widehat{S}(x) - S(x)| \le \exp\big(\widehat{H}(a) + \zeta + 1\big)\zeta.$$

W e prove the stronger bound

$$\sup_{0 \le x \le a} |\widehat{S}(x) - S(x)| \le \exp\{\widehat{H}(a)\}\zeta.$$

The stated inequality follows because

$$\exp\{\widehat{H}(a)\} \le \exp\{\widehat{H}(a) + \zeta + 1\}.$$

First suppose that both hazard measures are step functions with jumps at the finite ordered set

$$0 \le u_1 < \cdots < u_J \le a.$$

Include a point u_j whenever either H or $\widehat{H}$ has a jump there. Write

$$h_j := \Delta H(u_j), \qquad \hat{h}_j := \Delta \widehat{H}(u_j), \qquad \delta_j := \hat{h}_j - h_j.$$

Also define cumulative hazard differences

$$G_j := \sum_{k=1}^{j} \delta_k = \widehat{H}(u_j) - H(u_j).$$

By assumption,

$$|G_j| \le \zeta \qquad \text{for all } j = 1, \dots, J. \tag{35}$$

Let

$$S_0 = \widehat{S}_0 := 1,$$

and for $j \ge 1$,

$$S_j := \prod_{k=1}^{j} (1 - h_k), \qquad \widehat{S}_j := \prod_{k=1}^{j} (1 - \hat{h}_k).$$

Equivalently,

$$S_j = S_{j-1}(1 - h_j), \qquad \widehat{S}_j = \widehat{S}_{j-1}(1 - \hat{h}_j).$$

Since $h_j, \hat{h}_j \in [0,1]$, both sequences S_j and $\widehat{S}_j$ are nonincreasing and take values in $[0,1]$.

Define the survival difference

$$D_j := \widehat{S}_j - S_j.$$

Using the recursive equations,

$$\widehat{S}_j = 1 - \sum_{k=1}^{j} \widehat{S}_{k-1} \hat{h}_k, \qquad S_j = 1 - \sum_{k=1}^{j} S_{k-1} h_k.$$

Subtracting gives

$$\begin{aligned} D_j &= -\sum_{k=1}^{j} \left(\widehat{S}_{k-1} \hat{h}_k - S_{k-1} h_k \right) \\ &= -\sum_{k=1}^{j} D_{k-1} \hat{h}_k - \sum_{k=1}^{j} S_{k-1} (\hat{h}_k - h_k). \end{aligned} \tag{36}$$

The first term propagates previous survival errors through the estimated hazard. The second term is the direct effect of perturbing the hazard. We now bound the second term using Abel summation.

Set

$$B_j := \sum_{k=1}^{j} S_{k-1} \delta_k.$$

Since $G_k = \sum_{\ell=1}^{k} \delta_\ell$, Abel summation gives

$$B_j = S_{j-1} G_j + \sum_{k=1}^{j-1} (S_{k-1} - S_k) G_k. \tag{37}$$

Because S_k is nonincreasing and $0 \le S_k \le 1$,

$$S_{j-1} + \sum_{k=1}^{j-1} (S_{k-1} - S_k) = S_0 = 1.$$

Combining this identity with (35) and (37) yields

$$|B_j| \le \zeta \qquad \text{for all } j. \tag{38}$$

Now let

$$M_j := \max_{0 \le r \le j} |D_r|.$$

From (36) and (38),

$$|D_j| \le \zeta + \sum_{k=1}^{j} |D_{k-1}| \hat{h}_k \le \zeta + \sum_{k=1}^{j} M_{k-1} \hat{h}_k.$$

Hence

$$M_j \le \zeta + \sum_{k=1}^{j} M_{k-1} \hat{h}_k.$$

The discrete Gronwall inequality gives

$$M_j \le \zeta \prod_{k=1}^{j} (1 + \hat{h}_k).$$

Since $1 + x \le e^x$ for $x \ge 0$,

$$M_j \le \zeta \exp\left(\sum_{k=1}^{j} \hat{h}_k \right) \le \zeta e^{\widehat{H}(a)}.$$

Taking the supremum over j proves the desired bound for finite step hazards.

It remains to pass to general cumulative hazard measures. The product-integral survival functions satisfy the Stieltjes equations

$$S(x) = 1 - \int_{[0,x]} S(u-)\, dH(u), \qquad \widehat{S}(x) = 1 - \int_{[0,x]} \widehat{S}(u-)\, d\widehat{H}(u).$$

The same argument above can be written directly in Stieltjes form. Let

$$G(x) := \widehat{H}(x) - H(x), \qquad \|G\|_\infty \le \zeta.$$

Subtracting the two Stieltjes equations gives

$$\widehat{S}(x) - S(x) = -\int_{[0,x]} (\widehat{S}(u-) - S(u-))\, d\widehat{H}(u) - \int_{[0,x]} S(u-)\, dG(u).$$

The second integral is controlled by integration by parts. Since S is nonincreasing, takes values in $[0,1]$, and has total variation at most one on $[0,x]$,

$$\sup_{0 \le y \le x} \left| \int_{[0,y]} S(u-)\, dG(u) \right| \le \zeta.$$

Therefore, with

$$M(x) := \sup_{0 \le y \le x} |\widehat{S}(y) - S(y)|,$$

we have

$$M(x) \le \zeta + \int_{[0,x]} M(u-)\, d\widehat{H}(u).$$

The Stieltjes version of Gronwall's inequality [Andersen et al., 2012] yields

$$M(x) \leq \zeta \prod_{u \leq x} (1 + d\widehat{H}(u)) \leq \zeta e^{\widehat{H}(x)} \leq \zeta e^{\widehat{H}(a)}.$$

Thus

$$\sup_{0 \leq x \leq a} |\widehat{S}(x) - S(x)| \leq \zeta e^{\widehat{H}(a)} \leq \exp\bigl(\widehat{H}(a) + \zeta + 1\bigr)\zeta,$$

which proves the lemma.

B.3. Unified Kaplan–Meier certificate

Lemma 6 (Validity of the unified Kaplan–Meier certificate) *Under Assumptions 1–4, with probability at least* $1 - T^{-4}$*, the following holds simultaneously for all* $m \leq T$ *and all* $a \in \mathcal{A}_T$*:*

$$\left| \widehat{\mu}_{\mathrm{KM}}(m; a) - \int_0^a \mathbb{P}(\rho_t > x)\, dx \right| \leq e_{\mathrm{KM}}(m; a).$$

Consequently,

$$\mu_\rho \in \mathcal{C}_{\mathrm{KM}}(m) \qquad \textit{for all } m \leq T.$$

T he proof is the predictable-censoring analogue of classical Kaplan–Meier exponential inequalities. In the standard i.i.d. random-censoring model, Bitouzé et al. [1999] prove a Dvoretzky–Kiefer–Wolfowitz-type inequality for the Kaplan–Meier estimator. Here the censoring distribution is not i.i.d. because the censoring threshold is generated by the predictable episode-start order. The proof below replaces the i.i.d. censoring assumption by the conditional independence structure generated by our inventory model.

The proof uses only the episode-start observations in the unified shrinkage log

$$\mathscr{D}_\rho(m) = \{(U_\ell, Z_{t_\ell}) : 1 \leq \ell \leq m\}.$$

For episode ℓ, define

$$T_\ell := \rho_{t_\ell}, \qquad C_\ell := \frac{D_{t_\ell}}{\xi_\ell}.$$

The episode-start order ξ_ℓ is predictable at the beginning of the episode, while D_{t_ℓ} and ρ_{t_ℓ} are fresh and independent of the past. Hence

$$U_\ell = \min\{T_\ell, C_\ell\}.$$

Moreover,

$$Z_{t_\ell} = 1 \quad \Longleftrightarrow \quad T_\ell < C_\ell,$$

up to the censoring-tie event $T_\ell = C_\ell$. Conditional on ξ_ℓ and T_ℓ, the tie event is

$$D_{t_\ell} = \xi_\ell T_\ell,$$

which has probability zero under Assumption 4. Thus the unified log is a predictably generated right-censored sample with conditionally independent censoring.

Let

$$\bar{F}_\rho(x) := \mathbb{P}(\rho_t > x), \qquad \bar{F}_\rho^-(x) := \mathbb{P}(\rho_t \geq x).$$

Define the shrinkage hazard measure by

$$d\mathsf{H}_\rho(x) := \frac{\mathbb{P}(\rho_t \in dx)}{\bar{F}_\rho^-(x)}$$

on the set where $\bar{F}_\rho^-(x) > 0$. At an atom x, the jump $\Delta\mathsf{H}_\rho(x)$ is

$$\mathbb{P}(\rho_t = x \mid \rho_t \geq x).$$

The survival function is the product integral generated by this hazard:

$$\bar{F}_\rho(x) = \prod_{u \leq x} \big(1 - d\mathsf{H}_\rho(u)\big).$$

The value at $x = 0$ is immaterial for the restricted-mean integral.

For $x \in [0,1]$, define the pooled risk and event-counting processes

$$R_m(x) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \geq x\}, \qquad N_m(x) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \leq x,\ Z_{t_\ell} = 1\}.$$

The Kaplan–Meier estimator in (17) is the product-limit estimator associated with these processes, and

$$\widehat{\mathsf{H}}_m(x) := \int_{[0,x]} \frac{dN_m(u)}{R_m(u)} = \sum_{j: r_{j,m} \leq x} \frac{d_{j,m}}{n_{j,m}}$$

is its empirical hazard score.

We first verify the compensator carefully. For each episode-start observation ℓ, define the at-risk indicator and individual event-counting process

$$\mathsf{R}_\ell(u) := \mathbf{1}\{U_\ell \geq u\}, \qquad N_\ell(x) := \mathbf{1}\{U_\ell \leq x,\ Z_{t_\ell} = 1\}.$$

Thus

$$R_m(u) = \sum_{\ell=1}^{m} \mathsf{R}_\ell(u), \qquad N_m(x) = \sum_{\ell=1}^{m} N_\ell(x).$$

The filtration used here is the product-limit filtration in the level variable u, not the chronological inventory filtration. Informally, we reveal the censored sample in increasing order of the level u. Just before level u, the event $\mathsf{R}_\ell(u) = 1$ is known: observation ℓ is still at risk if and only if

$$U_\ell \geq u \quad \Longleftrightarrow \quad T_\ell \geq u \text{ and } C_\ell \geq u.$$

Hence $\mathsf{R}_\ell(u)$ is predictable with respect to the product-limit filtration.

Conditional on the episode-start history, the order ξ_ℓ is fixed, while D_{t_ℓ} and ρ_{t_ℓ} are fresh and independent. Therefore T_ℓ and C_ℓ are conditionally independent. The distribution of C_ℓ may depend on ξ_ℓ, and hence may vary across episodes, but this does not affect the argument below because only conditional independence is used.

On the event $\mathsf{R}_\ell(u) = 1$, we know both $T_\ell \geq u$ and $C_\ell \geq u$. The censoring information $C_\ell \geq u$ does not change the conditional law of T_ℓ beyond the fact that $T_\ell \geq u$. Therefore

$$\mathbb{P}(T_\ell \in du \mid T_\ell \geq u,\ C_\ell \geq u,\ \text{episode-start history}) = \mathbb{P}(T_\ell \in du \mid T_\ell \geq u) = d\mathsf{H}_\rho(u),$$

where

$$d\mathsf{H}_\rho(u) := \frac{\mathbb{P}(\rho_t \in du)}{\mathbb{P}(\rho_t \geq u)}.$$

At an atom u, this reads

$$\Delta\mathsf{H}_\rho(u) = \mathbb{P}(\rho_t = u \mid \rho_t \geq u).$$

If $T_\ell = u$ and $\mathsf{R}_\ell(u) = 1$, then $C_\ell \geq u$. The demand anti-concentration assumption rules out the censoring tie $C_\ell = u$, so $C_\ell > u$ almost surely and the event at u is observed as an uncensored event.

Consequently, the conditional event increment of observation ℓ satisfies

$$\mathbb{E}[dN_\ell(u) \mid \mathscr{F}^{(m)}_{u-}] = \mathsf{R}_\ell(u)\, d\mathsf{H}_\rho(u),$$

where $\mathscr{F}^{(m)}_{u-}$ denotes the product-limit filtration just before level u. Summing over $\ell = 1, \ldots, m$, we obtain

$$\mathbb{E}[dN_m(u) \mid \mathscr{F}^{(m)}_{u-}] = R_m(u)\, d\mathsf{H}_\rho(u).$$

Equivalently, the signed increment

$$dM_m(u) := dN_m(u) - R_m(u)\, d\mathsf{H}_\rho(u)$$

has zero conditional mean. Therefore

$$M_m(x) := N_m(x) - \int_{[0,x]} R_m(u)\, d\mathsf{H}_\rho(u) = \int_{[0,x]} dM_m(u)$$

is a martingale in the level variable x with respect to the product-limit filtration.

We now control

$$\widehat{\mathsf{H}}_m(x) - \mathsf{H}_\rho(x) = \int_{[0,x]} \frac{dM_m(u)}{R_m(u)}.$$

The proof must handle two issues: the risk denominator $R_m(a)$ is random, and the shrinkage distribution may have atoms. We handle the first issue by localizing at deterministic risk levels, and the second by refining the filtration at event atoms.

Fix $m \leq T$ and $a \in \mathcal{A}_T$. If $R_m(a) = 0$, then $e_{\mathrm{KM}}(m; a) = 1$, and the restricted-mean error is trivially bounded by one. Hence assume $R_m(a) \geq 1$.

For a deterministic integer $r \in \{1, \ldots, m\}$, define the stopping level

$$\tau_r := \inf\{x \in [0,a] : R_m(x) < r\},$$

with the convention $\tau_r = a$ if the set is empty. Consider the stopped hazard martingale

$$Z_{m,r}(x) := \int_{[0, x\wedge\tau_r]} \frac{dM_m(u)}{R_m(u)}, \qquad 0 \leq x \leq a.$$

On the stopped interval, the risk set is at least r.

At an atom u, we reveal the at-risk individuals one by one. Conditional on the risk set at u, each at-risk individual has event probability

$$\Delta\mathsf{H}_\rho(u) = \mathbb{P}(\rho_t = u \mid \rho_t \geq u),$$

because censoring is conditionally independent of shrinkage. The grouped event jump can therefore be written as a sum of individual martingale increments

$$\frac{B_k - \Delta\mathsf{H}_\rho(u)}{R_m(u)}.$$

Each refined increment is bounded in absolute value by $1/R_m(u) \le 1/r$. The same bound holds for increments in the continuous part. Hence the refined stopped martingale has jumps bounded by $1/r$.

Its predictable quadratic variation satisfies

$$\langle Z_{m,r}\rangle_a \le \int_{[0,a]} \frac{d\mathsf{H}_\rho(u)}{r} = \frac{\mathsf{H}_\rho(a)}{r}.$$

Freedman's maximal inequality, applied with the deterministic jump bound $1/r$, gives

$$\mathbb{P}\left(\sup_{0\le x\le a} |Z_{m,r}(x)| > \sqrt{\frac{2\mathsf{H}_\rho(a)y}{r}} + \frac{2y}{3r}\right) \le 2e^{-y}.$$

Set

$$y = \ell_m^{\mathrm{KM}} := \log\left(\frac{32(m\vee 1)|\mathcal{A}_T|}{\delta_m}\right).$$

A union bound over $r = 1,\dots,m$ shows that, with probability at least $1-\delta_m/(16|\mathcal{A}_T|)$, the preceding display holds simultaneously for all deterministic risk levels r.

On this simultaneous event, choose the realized value $r = R_m(a)$. Since $R_m(u) \ge R_m(a)$ for all $u \le a$, the stopping level satisfies $\tau_{R_m(a)} = a$. Therefore

$$\eta_{m,a} := \sup_{0\le x\le a} \left|\widehat{\mathsf{H}}_m(x) - \mathsf{H}_\rho(x)\right|$$

satisfies

$$\eta_{m,a} \le \sqrt{\frac{2\mathsf{H}_\rho(a)\ell_m^{\mathrm{KM}}}{R_m(a)}} + \frac{2\ell_m^{\mathrm{KM}}}{3R_m(a)}. \tag{39}$$

The bound in (39) contains the unknown $\mathsf{H}_\rho(a)$. Since

$$\mathsf{H}_\rho(a) \le \widehat{\mathsf{H}}_m(a) + \eta_{m,a},$$

(39) implies

$$\eta_{m,a} \le \sqrt{\frac{2(\widehat{\mathsf{H}}_m(a)+\eta_{m,a})\ell_m^{\mathrm{KM}}}{R_m(a)}} + \frac{2\ell_m^{\mathrm{KM}}}{3R_m(a)}.$$

Solving this quadratic inequality gives the conservative self-normalized bound

$$\eta_{m,a} \le 2\sqrt{\frac{(\widehat{\mathsf{H}}_m(a)+1)\ell_m^{\mathrm{KM}}}{R_m(a)}} + 8\frac{\ell_m^{\mathrm{KM}}}{R_m(a)} = \zeta_m(a). \tag{40}$$

By Lemma 5, applied with

$$H = \mathsf{H}_\rho, \qquad \widehat{H} = \widehat{\mathsf{H}}_m, \qquad \zeta = \zeta_m(a),$$

we obtain

$$\sup_{0\le x\le a} \left|\widehat{\bar{F}}_{\rho,m}(x) - \bar{F}_\rho(x)\right| \le \exp\big(\widehat{\mathsf{H}}_m(a)+\zeta_m(a)+1\big)\zeta_m(a).$$

By the definition of $e_{\mathrm{KM}}(m;a)$, the right-hand side is at most $e_{\mathrm{KM}}(m;a)$. Since $a \le 1$, integration over $x\in[0,a]$ gives

$$\left|\widehat{\mu}_{\mathrm{KM}}(m;a) - \int_0^a \mathbb{P}(\rho_t > x)\,dx\right| \le e_{\mathrm{KM}}(m;a).$$

A union bound over $a\in\mathcal{A}_T$ and $m \le T$, using the summable schedule

$$\delta_m = \frac{1}{64T^4(m+1)^2},$$

shows that the restricted-mean inequality holds simultaneously for all $m \le T$ and all $a \in \mathcal{A}_T$ with probability at least $1 - T^{-4}$.

Finally,

$$\mu_\rho = \int_0^1 \mathbb{P}(\rho_t > x)\,dx = \int_0^a \mathbb{P}(\rho_t > x)\,dx + \int_a^1 \mathbb{P}(\rho_t > x)\,dx,$$

and

$$0 \le \int_a^1 \mathbb{P}(\rho_t > x)\,dx \le 1 - a.$$

Therefore

$$\mu_\rho \in \mathcal{C}_{\mathrm{KM}}(m; a)$$

for every $a \in \mathcal{A}_T$. In particular, this holds for the selected a_m^{KM}, and hence

$$\mu_\rho \in \mathcal{C}_{\mathrm{KM}}(m) \qquad \text{for all } m \le T.$$

This proves Lemma 6.

B.4. Cycle concentration and implementation stability

Lemma 7 (Uniform cycle concentration) *With probability at least $1 - 2T^{-4}$, the following holds simultaneously for all arms $i \in \{1, \dots, M\}$ and all $m \le T$ with $N_i(m) \ge 1$:*

$$\left| \frac{\sum_{\ell \in \mathcal{I}_i(m)} \left(\widetilde{K}_\ell^{\mathrm{obs}}(\mu_\rho) - \gamma(s_i; \nu_{\ell-1}) \Lambda_\ell \right)}{\sum_{\ell \in \mathcal{I}_i(m)} \Lambda_\ell} \right| \le B_i^{\mathrm{stat}}(m).$$

W ork on the event from Lemma 2 that all episodes initiated before T have length at most $\bar{\Lambda}_T$. Fix an arm i and enumerate the completed decision episodes of this arm by

$$\ell_{i,1} < \ell_{i,2} < \cdots.$$

For the jth such episode, write

$$\Lambda_{i,j} := \Lambda_{\ell_{i,j}}, \qquad \widetilde{K}_{i,j}^{\mathrm{obs}}(\mu) := \widetilde{K}_{\ell_{i,j}}^{\mathrm{obs}}(\mu), \qquad \nu_{i,j} := \nu_{\ell_{i,j}-1}.$$

Define

$$X_{i,j} := \widetilde{K}_{i,j}^{\mathrm{obs}}(\mu_\rho) - \gamma(s_i; \nu_{i,j}) \Lambda_{i,j}.$$

Conditional on the history up to the start of episode $\ell_{i,j}$, the fresh exogenous sequence is independent, the episode starts from the stockout state, and the chosen pair $(s_i, \nu_{i,j})$ is predictable. Hence

$$\mathbb{E}[X_{i,j} \mid \mathcal{G}_{\ell_{i,j}-1}] = 0.$$

Moreover,

$$|X_{i,j}| \le |\widetilde{K}_{i,j}^{\mathrm{obs}}(\mu_\rho)| + |\gamma(s_i; \nu_{i,j})| \Lambda_{i,j} \le 2\bar{c}_{\mathrm{per}} \bar{\Lambda}_T.$$

Azuma–Hoeffding therefore yields, for every fixed $n \ge 1$,

$$\mathbb{P}\left(\left| \sum_{j=1}^n X_{i,j} \right| > 2\sqrt{2}\,\bar{c}_{\mathrm{per}} \bar{\Lambda}_T \sqrt{n L_{M,T}} \right) \le 2e^{-L_{M,T}}.$$

Because $\sum_{j=1}^n \Lambda_{i,j} \ge n$, dividing by the denominator gives the announced bound. A union bound over $i = 1, \dots, M$ and $n = 1, \dots, T$, together with the failure probability from Lemma 2, yields the result.

Lemma 8 (Implementation stability) *Under Assumptions 1, 3, and 4, there exists a finite constant L_{impl}, depending only on the model primitives, such that for every $S \in [0, \bar{S}]$ and every $\nu, \nu' \in [\underline{\mu}, \bar{\mu}]$,*

$$|g(S;\nu) - g(S;\nu')| \le L_{\mathrm{impl}} |\nu - \nu'|.$$

Equivalently,

$$|\gamma(S;\nu) - \gamma(S;\nu')| \le L_{\mathrm{impl}} |\nu - \nu'|.$$

F ix $S \in [0, \bar{S}]$ and $\nu, \nu' \in [\underline{\mu}, \bar{\mu}]$. The comparison is made over one generic regenerative cycle. Couple the two systems with the same exogenous sequence $(D_t, \rho_t)_{t \ge 1}$, and start both systems from a stockout state:

$$X_1^{\nu} = X_1^{\nu'} = 0, \qquad \hat{X}_1^{\nu} = \hat{X}_1^{\nu'} = 0.$$

All time indices in this proof are local to this fresh cycle. Let

$$\Delta_{\nu} := |\nu - \nu'|.$$

Define the two cycle lengths by

$$\Lambda^{\nu} := \inf\{t \ge 1 : Z_t^{\nu} = 1\}, \qquad \Lambda^{\nu'} := \inf\{t \ge 1 : Z_t^{\nu'} = 1\}.$$

Thus, if the two systems stock out together in period s, then $\Lambda^{\nu} = \Lambda^{\nu'} = s$, and the coupled-cycle comparison stops there. Define the first stockout-pattern mismatch by

$$M := \inf\{t \ge 1 : \Lambda^{\nu} \ge t,\ \Lambda^{\nu'} \ge t,\ Z_t^{\nu} \ne Z_t^{\nu'}\},$$

with $M = \infty$ if no such period occurs. For $t \ge 1$, define the event that both coupled cycles are alive at the start of local period t:

$$\mathcal{H}_t := \{\Lambda^{\nu} \ge t,\ \Lambda^{\nu'} \ge t\}. \tag{41}$$

On $\mathcal{H}_t$, neither system has stocked out in periods $1, \ldots, t-1$.

We first record pathwise sensitivity bounds. For every $r \le t$, on $\mathcal{H}_t$,

$$|\hat{X}_r^{\nu} - \hat{X}_r^{\nu'}| \le S\Delta_{\nu}, \qquad |q_r^{\nu} - q_r^{\nu'}| \le S\Delta_{\nu}, \tag{42}$$

and

$$|X_r^{\nu} - X_r^{\nu'}| \le (r-1)S\Delta_{\nu}, \qquad |A_r^{\nu} - A_r^{\nu'}| \le rS\Delta_{\nu}. \tag{43}$$

Indeed, within a cycle the tracker satisfies $\hat{X}_r \le S$, so $q_r = S - \hat{X}_r$. On $\mathcal{H}_t$, there were no stockouts in periods $1, \ldots, t-1$. Therefore, for every preceding period $r < t$,

$$\hat{X}_{r+1}^{\nu} = (\nu S - D_r)^+, \qquad \hat{X}_{r+1}^{\nu'} = (\nu' S - D_r)^+.$$

Since $x \mapsto x^+$ is one-Lipschitz,

$$|\hat{X}_{r+1}^{\nu} - \hat{X}_{r+1}^{\nu'}| \le S\Delta_{\nu}.$$

The order bound in (42) follows from $q_r = S - \hat{X}_r$. For the true inventories, whenever the two cycles are alive and no stockout occurs,

$$X_{r+1}^{\nu} = \rho_r(X_r^{\nu} + q_r^{\nu}) - D_r, \qquad X_{r+1}^{\nu'} = \rho_r(X_r^{\nu'} + q_r^{\nu'}) - D_r.$$

Hence, using $\rho_r \le 1$,

$$|X_{r+1}^{\nu} - X_{r+1}^{\nu'}| \le |X_r^{\nu} - X_r^{\nu'}| + |q_r^{\nu} - q_r^{\nu'}|.$$

Starting from $X_1^{\nu} = X_1^{\nu'} = 0$, induction gives the true-inventory bound in (43). Since

$$A_r^{\nu} = \rho_r(X_r^{\nu} + q_r^{\nu}), \qquad A_r^{\nu'} = \rho_r(X_r^{\nu'} + q_r^{\nu'}),$$

the bound on usable inventory follows from the true-inventory and order bounds.

If a stockout-pattern mismatch occurs in period t, then both cycles are alive at the start of that period, and the realized demand must fall between the two usable-inventory thresholds. Let $\mathcal{G}_{t-1}$ denote the history up to the start of local period t. Using Assumption 4 and (43),

$$\begin{aligned}\mathbb{P}(M = t) &\le \mathbb{E}\Big[\mathbf{1}_{\mathcal{H}_t}\mathbb{P}\Big(D_t \in [A_t^{\nu} \wedge A_t^{\nu'},\, A_t^{\nu} \vee A_t^{\nu'}] \,|\, \mathcal{G}_{t-1}, \rho_t\Big)\Big] \\ &\le \bar{f}_D\, tS\Delta_\nu\, \mathbb{P}(\mathcal{H}_t). \end{aligned} \tag{44}$$

If $\mathcal{H}_t$ occurs, neither system has stocked out in periods $1, \dots, t-1$. Since $\rho_s = 0$ forces a stockout in period s, and $\mathbb{P}(\rho_s = 0) \ge p_0$,

$$\mathbb{P}(\mathcal{H}_t) \le (1 - p_0)^{t-1}. \tag{45}$$

Combining (44) and (45) gives

$$\mathbb{P}(M = t) \le \bar{f}_D\, tS\Delta_\nu(1 - p_0)^{t-1}. \tag{46}$$

Consequently,

$$\mathbb{P}(M < \infty) \le \sum_{t \ge 1} \bar{f}_D\, tS\Delta_\nu(1 - p_0)^{t-1} = \frac{\bar{f}_D S}{p_0^2}\Delta_\nu \le \frac{\bar{f}_D \bar{S}}{p_0^2}\Delta_\nu. \tag{47}$$

Let

$$\alpha_{\max} := \alpha(\bar{\mu}), \qquad \bar{c}_{\text{per}} := \alpha_{\max}\bar{S} + (\alpha_{\max} + h + b)\overline{D},$$

and define the one-period pseudo-cost increments

$$k_t^{\nu} := \alpha(\mu_\rho)(q_t^{\nu} - Y_t^{\nu}) - (h + b)Y_t^{\nu}, \qquad k_t^{\nu'} := \alpha(\mu_\rho)(q_t^{\nu'} - Y_t^{\nu'}) - (h + b)Y_t^{\nu'}.$$

The corresponding cycle pseudo-costs are

$$\widetilde{K}^{S,\nu}(\mu_\rho) := \sum_{t=1}^{\Lambda^{\nu}} k_t^{\nu}, \qquad \widetilde{K}^{S,\nu'}(\mu_\rho) := \sum_{t=1}^{\Lambda^{\nu'}} k_t^{\nu'}.$$

The sales map is one-Lipschitz in usable inventory. For every common demand realization D_t,

$$|Y_t^{\nu} - Y_t^{\nu'}| = \Big|\min\{A_t^{\nu}, D_t\} - \min\{A_t^{\nu'}, D_t\}\Big| \le |A_t^{\nu} - A_t^{\nu'}|. \tag{48}$$

This inequality holds whether neither system stocks out, both systems stock out, or exactly one system stocks out in period t. Combining (48) with (42) and (43), we obtain, on $\mathcal{H}_t$,

$$\begin{aligned}|k_t^{\nu} - k_t^{\nu'}| &\le \alpha_{\max}|q_t^{\nu} - q_t^{\nu'}| + (\alpha_{\max} + h + b)|Y_t^{\nu} - Y_t^{\nu'}| \\ &\le \alpha_{\max} S\Delta_\nu + (\alpha_{\max} + h + b)tS\Delta_\nu \\ &\le c_1 t\Delta_\nu, \end{aligned} \tag{49}$$

where

$$c_1 := (2\alpha_{\max} + h + b)\bar{S}.$$

The expected contribution from periods in which both coupled cycles are alive is therefore bounded by

$$\begin{aligned}\mathbb{E}\left[\sum_{t\geq 1} |k_t^{\nu} - k_t^{\nu'}|\mathbf{1}_{\mathcal{H}_t}\right] &\leq \sum_{t\geq 1} c_1 t \Delta_\nu \, \mathbb{P}(\mathcal{H}_t) \\ &\leq \sum_{t\geq 1} c_1 t \Delta_\nu (1-p_0)^{t-1} \\ &= \frac{c_1}{p_0^2}\Delta_\nu. \end{aligned} \tag{50}$$

If a stockout-pattern mismatch occurs, then after the mismatch period one of the two coupled cycles may continue while the other has ended. Each one-period pseudo-cost is bounded in absolute value by $\bar{c}_{\text{per}}$, and, conditional on the mismatch history, the expected remaining length of any still-alive cycle is at most $1/p_0$. Therefore, using (47),

$$\mathbb{E}[\text{post-mismatch pseudo-cost difference}] \leq \frac{2\bar{c}_{\text{per}}}{p_0}\mathbb{P}(M < \infty) \leq \frac{2\bar{c}_{\text{per}}\bar{f}_D\bar{S}}{p_0^3}\Delta_\nu. \tag{51}$$

Equations (50) and (51) imply

$$\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)] - \mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right| \leq L_K \Delta_\nu, \tag{52}$$

where one may take

$$L_K := \frac{c_1}{p_0^2} + \frac{2\bar{c}_{\text{per}}\bar{f}_D\bar{S}}{p_0^3}. \tag{53}$$

The expected cycle lengths are also Lipschitz. If the two systems stock out together, their cycle lengths are equal. The lengths can differ only if a stockout-pattern mismatch occurs. Conditional on such a mismatch, the expected remaining length of any still-alive cycle is at most $1/p_0$. Therefore, using (47),

$$\left|\mathbb{E}[\Lambda^\nu] - \mathbb{E}[\Lambda^{\nu'}]\right| \leq \frac{2}{p_0}\mathbb{P}(M < \infty) \leq \frac{2\bar{f}_D\bar{S}}{p_0^3}\Delta_\nu =: L_\Lambda \Delta_\nu. \tag{54}$$

The pseudo-cost rates are

$$\gamma(S;\nu) = \frac{\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda^\nu]}, \qquad \gamma(S;\nu') = \frac{\mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]}{\mathbb{E}[\Lambda^{\nu'}]}.$$

Since every cycle lasts at least one period,

$$\mathbb{E}[\Lambda^\nu] \geq 1, \qquad \mathbb{E}[\Lambda^{\nu'}] \geq 1.$$

Moreover, by the forced-regeneration condition and the envelope $\bar{c}_{\text{per}}$,

$$\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]\right| \leq \frac{\bar{c}_{\text{per}}}{p_0}, \qquad \left|\mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right| \leq \frac{\bar{c}_{\text{per}}}{p_0}. \tag{55}$$

Using (52)–(55),

$$\begin{aligned}|\gamma(S;\nu) - \gamma(S;\nu')| &\leq \left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)] - \mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right| \\ &\quad + \left|\mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right| \left|\frac{1}{\mathbb{E}[\Lambda^\nu]} - \frac{1}{\mathbb{E}[\Lambda^{\nu'}]}\right| \\ &\leq L_K\Delta_\nu + \frac{\bar{c}_{\text{per}}}{p_0}L_\Lambda\Delta_\nu. \end{aligned} \tag{56}$$

Thus $\gamma(S;\nu)$ is Lipschitz in ν. Since

$$g(S;\nu) = b\mathbb{E}[D_t] + \gamma(S;\nu),$$

the same Lipschitz bound holds for $g(S;\nu)$. Taking

$$L_{\text{impl}} := L_K + \frac{\bar{c}_{\text{per}}}{p_0}L_\Lambda$$

completes the proof.

B.5. Arm confidence and optimism

Lemma 9 (Uniform confidence for arm pseudo-cost rates) *For a fixed grid, let*

$$i^\star \in \arg\min_{1\le i\le M} g(s_i), \qquad S_M^\star := s_{i^\star}.$$

Set, by convention,

$$\mathcal{C}_\mu(0) := [\underline{\mu}, \bar{\mu}], \qquad r_\mu(0) := \frac{\bar{\mu}-\underline{\mu}}{2}.$$

There exists an event $\mathcal{E}_T$ with probability at least $1-6T^{-4}$ such that, on $\mathcal{E}_T$, the following statements hold.

(i) For every $0\le m\le T$, the shrinkage confidence interval is valid:

$$\mu_\rho \in \mathcal{C}_\mu(m).$$

(ii) For every $m\le T$ and every arm i with $N_i(m)\ge 1$, the empirical arm score is accurate:

$$|\hat{\gamma}_i(m) - \gamma(s_i;\mu_\rho)| \le B_i(m).$$

(iii) The grid-oracle arm $i^\star$ is never eliminated.

(iv) Whenever episode $m+1$ is a decision episode and the chosen arm i_m has already been sampled, we have

$$g(s_{i_m}) - g(S_M^\star) \le 2B_{i_m}(m).$$

L et $\mathcal{E}_T$ be the intersection of the following events:

(a) all Kaplan–Meier certificates are valid up to episode T, as in Lemma 6;

(b) all bracketing certificates are valid up to episode T, as in Lemma 3;

(c) all probe certificates are valid up to episode T, as in Lemma 4;

(d) all cycle-concentration inequalities from Lemma 7 hold up to episode T;

(e) the bounded-length event from Lemma 2 holds for all episodes initiated before horizon T.

By Lemmas 6, 3, 4, 7, and 2, together with a union bound,

$$\mathbb{P}(\mathcal{E}_T^c) \le T^{-4} + T^{-4} + T^{-4} + 2T^{-4} + T^{-4} = 6T^{-4}.$$

Hence

$$\mathbb{P}(\mathcal{E}_T) \ge 1 - 6T^{-4}.$$

On $\mathcal{E}_T$, the Kaplan–Meier, bracketing, and probe certificates all contain μ_ρ. Therefore their intersection $\mathcal{C}_\mu(m)$ is nonempty and contains μ_ρ for every $m\le T$. Since $\hat{\mu}_m$ is the midpoint of this interval,

$$|\hat{\mu}_m - \mu_\rho| \le r_\mu(m), \qquad 0\le m\le T.$$

This proves part (i).

Fix $m\le T$ and an arm i with $N_i(m)\ge 1$. Write

$$\gamma_i := \gamma(s_i;\mu_\rho), \qquad \mathcal{T}_i(m) := \sum_{\ell\in\mathcal{I}_i(m)} \Lambda_\ell.$$

Using the realized pseudo-cost notation in (25), we have the exact decomposition

$$\hat{\gamma}_i(m) - \gamma_i = A_i^{\text{stat}}(m) + A_i^{\mu,\text{eval}}(m) + A_i^{\mu,\text{impl}}(m),$$

where

$$A_i^{\text{stat}}(m) := \frac{\sum_{\ell \in \mathcal{I}_i(m)} \left(\widetilde{K}_\ell^{\text{obs}}(\mu_\rho) - \gamma(s_i; \nu_{\ell-1})\Lambda_\ell\right)}{\mathcal{T}_i(m)},$$

$$A_i^{\mu,\text{eval}}(m) := \frac{\sum_{\ell \in \mathcal{I}_i(m)} [\alpha(\hat{\mu}_m) - \alpha(\mu_\rho)] W_\ell}{\mathcal{T}_i(m)},$$

and

$$A_i^{\mu,\text{impl}}(m) := \frac{\sum_{\ell \in \mathcal{I}_i(m)} [\gamma(s_i; \nu_{\ell-1}) - \gamma(s_i; \mu_\rho)]\Lambda_\ell}{\mathcal{T}_i(m)}.$$

By Lemma 7,

$$|A_i^{\text{stat}}(m)| \leq B_i^{\text{stat}}(m).$$

Since $|\hat{\mu}_m - \mu_\rho| \leq r_\mu(m)$, and since

$$0 \leq W_\ell \leq Q_\ell \leq \bar{S}\Lambda_\ell,$$

we have

$$|A_i^{\mu,\text{eval}}(m)| \leq L_\alpha \bar{S} r_\mu(m) = B_i^{\mu,\text{eval}}(m).$$

Finally, by Lemma 8,

$$|\gamma(s_i; \nu_{\ell-1}) - \gamma(s_i; \mu_\rho)| \leq L_{\text{impl}} |\nu_{\ell-1} - \mu_\rho|.$$

On $\mathcal{E}_T$,

$$|\nu_{\ell-1} - \mu_\rho| = |\hat{\mu}_{\ell-1} - \mu_\rho| \leq r_\mu(\ell - 1),$$

where the convention for $r_\mu(0)$ handles the first possible decision episode. Therefore

$$|A_i^{\mu,\text{impl}}(m)| \leq L_{\text{impl}} \frac{\sum_{\ell \in \mathcal{I}_i(m)} \Lambda_\ell r_\mu(\ell - 1)}{\sum_{\ell \in \mathcal{I}_i(m)} \Lambda_\ell} = B_i^{\mu,\text{impl}}(m).$$

Combining the three bounds gives

$$|\hat{\gamma}_i(m) - \gamma(s_i; \mu_\rho)| \leq B_i^{\text{stat}}(m) + B_i^{\mu,\text{eval}}(m) + B_i^{\mu,\text{impl}}(m) = B_i(m).$$

This proves part (ii).

We next prove that the grid-oracle arm is never eliminated. If $N_{i^\star}(m) = 0$, then by convention

$$\text{LCB}_{i^\star}(m) = -\infty,$$

so $i^\star$ cannot be removed by the elimination rule. Suppose instead that $N_{i^\star}(m) \geq 1$. Then part (ii) gives

$$\text{LCB}_{i^\star}(m) = \hat{\gamma}_{i^\star}(m) - B_{i^\star}(m) \leq \gamma(s_{i^\star}; \mu_\rho).$$

For any active arm j, if $N_j(m) = 0$, then

$$\text{UCB}_j(m) = +\infty,$$

and hence

$$\text{LCB}_{i^\star}(m) \leq \text{UCB}_j(m).$$

If $N_j(m) \geq 1$, then part (ii) gives

$$\mathrm{UCB}_j(m) = \hat{\gamma}_j(m) + B_j(m) \geq \gamma(s_j; \mu_\rho).$$

Since $i^\star$ minimizes $g(s_i)$ over the grid and

$$g(s_i) = b\mathbb{E}[D_t] + \gamma(s_i; \mu_\rho),$$

we have

$$\gamma(s_{i^\star}; \mu_\rho) \leq \gamma(s_j; \mu_\rho).$$

Therefore, for every active arm j,

$$\mathrm{LCB}_{i^\star}(m) \leq \mathrm{UCB}_j(m).$$

Thus the elimination condition can never remove $i^\star$. This proves part (iii).

Finally, suppose episode $m+1$ is a decision episode and the chosen arm i_m has already been sampled:

$$N_{i_m}(m) \geq 1.$$

By the algorithm's initialization rule, if some active arm were unsampled, the algorithm would sample an unsampled active arm before applying the optimistic selection rule. Therefore, in this case all active arms have been sampled. By part (iii), $i^\star$ is active, and hence

$$N_{i^\star}(m) \geq 1.$$

The optimistic rule gives

$$\mathrm{LCB}_{i_m}(m) \leq \mathrm{LCB}_{i^\star}(m).$$

Using part (ii),

$$\mathrm{LCB}_{i^\star}(m) \leq \gamma(s_{i^\star}; \mu_\rho),$$

and

$$\gamma(s_{i_m}; \mu_\rho) \leq \mathrm{UCB}_{i_m}(m) = \mathrm{LCB}_{i_m}(m) + 2B_{i_m}(m).$$

Combining these inequalities yields

$$\gamma(s_{i_m}; \mu_\rho) - \gamma(s_{i^\star}; \mu_\rho) \leq 2B_{i_m}(m).$$

Since the additive term $b\mathbb{E}[D_t]$ is common to all grid arms,

$$g(s_{i_m}) - g(S_M^\star) = \gamma(s_{i_m}; \mu_\rho) - \gamma(s_{i^\star}; \mu_\rho) \leq 2B_{i_m}(m).$$

This proves part (iv), and completes the proof.

B.6. Base-stock sensitivity and discretization

Lemma 10 (Base-stock stability) *Under Assumptions 1, 3, and 4, there exists a finite constant L_S, depending only on the model primitives, such that for every tracker parameter $\nu \in [\underline{\mu}, \bar{\mu}]$ and every $S, S' \in [0, \bar{S}]$,*

$$|g(S;\nu) - g(S';\nu)| \le L_S |S - S'|.$$

In particular,

$$|g(S) - g(S')| \le L_S |S - S'|.$$

F ix $\nu \in [\underline{\mu}, \bar{\mu}]$ and $S, S' \in [0, \bar{S}]$. Couple the two systems with the same exogenous sequence $(D_t, \rho_t)_{t \ge 1}$, and start both cycles from

$$X_1^S = X_1^{S'} = 0, \qquad \hat{X}_1^S = \hat{X}_1^{S'} = 0.$$

Write

$$\Delta_S := |S - S'|.$$

Let

$$\Lambda^S := \inf\{t \ge 1 : Z_t^S = 1\}, \qquad \Lambda^{S'} := \inf\{t \ge 1 : Z_t^{S'} = 1\}$$

be the two coupled cycle lengths, and let

$$M := \inf\{t \ge 1 : Z_t^S \ne Z_t^{S'}\}$$

be the first period in which the two coupled systems disagree on the stockout indicator. If no such disagreement occurs before both cycles end, set $M = \infty$. For $t \ge 1$, define

$$\mathcal{H}_t := \{\Lambda^S \ge t,\ \Lambda^{S'} \ge t,\ M \ge t\}. \tag{57}$$

Thus, on $\mathcal{H}_t$, both systems are alive at the start of period t, and their stockout histories agree through period $t-1$.

We first record the pathwise sensitivity of the coupled systems on $\mathcal{H}_t$. For every $r \le t$, on $\mathcal{H}_t$,

$$|\hat{X}_r^S - \hat{X}_r^{S'}| \le \Delta_S, \qquad |q_r^S - q_r^{S'}| \le 2\Delta_S, \tag{58}$$

and

$$|X_r^S - X_r^{S'}| \le 2(r-1)\Delta_S, \qquad |A_r^S - A_r^{S'}| \le 2r\Delta_S. \tag{59}$$

Indeed, the tracker always satisfies $\hat{X}_r^S \le S$ and $\hat{X}_r^{S'} \le S'$ within a cycle, so

$$q_r^S = S - \hat{X}_r^S, \qquad q_r^{S'} = S' - \hat{X}_r^{S'}.$$

On $\mathcal{H}_t$, the stockout histories coincide before period t. Hence in every preceding non-stockout period $r < t$,

$$\hat{X}_{r+1}^S = (\nu S - D_r)^+, \qquad \hat{X}_{r+1}^{S'} = (\nu S' - D_r)^+.$$

Since $x \mapsto x^+$ is one-Lipschitz and $\nu \le 1$,

$$|\hat{X}_{r+1}^S - \hat{X}_{r+1}^{S'}| \le \nu |S - S'| \le \Delta_S.$$

The order bound in (58) follows from $q_r = S - \hat{X}_r$. For the true inventories, whenever both coupled cycles are alive and no stockout occurs,

$$X^S_{r+1} = \rho_r(X^S_r + q^S_r) - D_r, \qquad X^{S'}_{r+1} = \rho_r(X^{S'}_r + q^{S'}_r) - D_r.$$

Therefore, using $\rho_r \le 1$,

$$|X^S_{r+1} - X^{S'}_{r+1}| \le |X^S_r - X^{S'}_r| + |q^S_r - q^{S'}_r|.$$

Starting from $X^S_1 = X^{S'}_1 = 0$ and using (58), induction gives the true-inventory bound in (59). Since

$$A^S_r = \rho_r(X^S_r + q^S_r), \qquad A^{S'}_r = \rho_r(X^{S'}_r + q^{S'}_r),$$

the bound on $|A^S_r - A^{S'}_r|$ follows as well.

A stockout-pattern mismatch in period t can occur only if both systems are alive at the start of period t, their stockout histories agree before that period, and D_t falls between the two usable-inventory thresholds. Hence, using Assumption 4 and (59),

$$\begin{aligned}\mathbb{P}(M = t) &\le \mathbb{E}\Big[\mathbf{1}_{\mathcal{H}_t}\mathbb{P}\Big(D_t \in [A^S_t \wedge A^{S'}_t,\, A^S_t \vee A^{S'}_t] \mid \mathcal{G}_{t-1}, \rho_t\Big)\Big] \\ &\le 2\bar{f}_D\, t\Delta_S\, \mathbb{P}(\mathcal{H}_t).\end{aligned} \tag{60}$$

If $\mathcal{H}_t$ occurs, then neither coupled cycle has stocked out in periods $1, \dots, t-1$. Since $\rho_s = 0$ forces a stockout in period s and $\mathbb{P}(\rho_s = 0) \ge p_0$,

$$\mathbb{P}(\mathcal{H}_t) \le (1 - p_0)^{t-1}. \tag{61}$$

Combining (60) and (61) gives

$$\mathbb{P}(M = t) \le 2\bar{f}_D\, t\Delta_S (1 - p_0)^{t-1}. \tag{62}$$

Therefore

$$\mathbb{P}(M < \infty) \le \sum_{t \ge 1} 2\bar{f}_D\, t\Delta_S (1 - p_0)^{t-1} = \frac{2\bar{f}_D}{p_0^2}\Delta_S. \tag{63}$$

Let

$$\alpha_{\max} := \alpha(\bar{\mu}), \qquad \bar{c}_{\text{per}} := \alpha_{\max}\bar{S} + (\alpha_{\max} + h + b)\overline{D},$$

and define the one-period pseudo-cost increment under base-stock level S by

$$k^S_t := \alpha(\mu_\rho)(q^S_t - Y^S_t) - (h + b)Y^S_t.$$

Let

$$\widetilde{K}^{S,\nu}(\mu_\rho) := \sum_{t=1}^{\Lambda^S} k^S_t$$

be the cycle pseudo-cost under base-stock level S and tracker parameter ν. Set

$$c_1 := 2(2\alpha_{\max} + h + b).$$

On the event $\mathcal{H}_t \cap \{M > t\}$, the two stockout indicators agree in period t. Hence

$$|Y^S_t - Y^{S'}_t| \le |A^S_t - A^{S'}_t|.$$

Using (58) and (59), we obtain

$$
\begin{aligned}
|k_t^S - k_t^{S'}| &\le \alpha_{\max}|q_t^S - q_t^{S'}| + (\alpha_{\max} + h + b)|Y_t^S - Y_t^{S'}| \\
&\le 2\alpha_{\max}\Delta_S + 2(\alpha_{\max} + h + b)t\Delta_S \\
&\le c_1 t\Delta_S. \qquad (64)
\end{aligned}
$$

The expected contribution before the first mismatch is therefore bounded by

$$
\begin{aligned}
\mathbb{E}\left[\sum_{t\ge 1} |k_t^S - k_t^{S'}|\mathbf{1}_{\mathcal{H}_t}\mathbf{1}\{M > t\}\right] &\le \sum_{t\ge 1} c_1 t\Delta_S(1-p_0)^{t-1} \\
&= \frac{c_1}{p_0^2}\Delta_S. \qquad (65)
\end{aligned}
$$

After the first mismatch, each one-period pseudo-cost is bounded in absolute value by $\bar{c}_{\text{per}}$, and conditional on a mismatch the expected remaining length of each coupled cycle is at most $1/p_0$. Hence

$$
\mathbb{E}[\text{post-mismatch pseudo-cost difference}] \le \frac{2\bar{c}_{\text{per}}}{p_0}\mathbb{P}(M < \infty) \le \frac{4\bar{c}_{\text{per}}\bar{f}_D}{p_0^3}\Delta_S. \qquad (66)
$$

Equations (65) and (66) imply

$$
\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)] - \mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right| \le L_K^S\Delta_S, \qquad (67)
$$

where one may take

$$
L_K^S := \frac{c_1}{p_0^2} + \frac{4\bar{c}_{\text{per}}\bar{f}_D}{p_0^3}. \qquad (68)
$$

The expected cycle lengths are also Lipschitz. If no stockout-pattern mismatch occurs, then the two coupled cycles have the same length. Conditional on a mismatch, the expected remaining length of each coupled cycle is at most $1/p_0$. Thus (63) gives

$$
\left|\mathbb{E}[\Lambda^S(\nu)] - \mathbb{E}[\Lambda^{S'}(\nu)]\right| \le \frac{2}{p_0}\mathbb{P}(M < \infty) \le \frac{4\bar{f}_D}{p_0^3}\Delta_S =: L_\Lambda^S\Delta_S. \qquad (69)
$$

The pseudo-cost rates are

$$
\gamma(S;\nu) = \frac{\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda^{S,\nu}]}, \qquad \gamma(S';\nu) = \frac{\mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda^{S',\nu}]}.
$$

Since every cycle lasts at least one period,

$$
\mathbb{E}[\Lambda^S(\nu)] \ge 1, \qquad \mathbb{E}[\Lambda^{S'}(\nu)] \ge 1.
$$

Moreover,

$$
\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]\right| \le \frac{\bar{c}_{\text{per}}}{p_0}, \qquad \left|\mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right| \le \frac{\bar{c}_{\text{per}}}{p_0}. \qquad (70)
$$

Using (67)–(70),

$$
\begin{aligned}
|\gamma(S;\nu) - \gamma(S';\nu)| &\le \left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)] - \mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right| \\
&\quad + \left|\mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right| \left|\frac{1}{\mathbb{E}[\Lambda^S(\nu)]} - \frac{1}{\mathbb{E}[\Lambda^{S'}(\nu)]}\right| \\
&\le L_K^S\Delta_S + \frac{\bar{c}_{\text{per}}}{p_0}L_\Lambda^S\Delta_S. \qquad (71)
\end{aligned}
$$

Thus $\gamma(S;\nu)$ is Lipschitz in S, uniformly over $\nu \in [\underline{\mu}, \bar{\mu}]$. Since

$$g(S;\nu) = b\mathbb{E}[D_t] + \gamma(S;\nu),$$

the same Lipschitz bound holds for $g(S;\nu)$. Taking

$$L_S := L_K^S + \frac{\bar{c}_{\mathrm{per}}}{p_0} L_\Lambda^S$$

proves the first claim. Setting $\nu = \mu_\rho$ gives

$$|g(S) - g(S')| \le L_S |S - S'|.$$

B.7. Pathwise summation lemmas

Threshold tradeoff. The shrinkage-learning part of the regret bound depends on two pathwise quantities:

$$P_T \qquad \text{and} \qquad \sum_{m \in \mathcal{D}_T} r_m^-.$$

The first is the number of clean probes, which is multiplied by the direct probe cost. The second is the cumulative shrinkage radius, which controls pseudo-cost evaluation error and tracker implementation error. The next two lemmas show that the threshold rule controls these two quantities separately. Together, they are the formal version of the audit–inaccuracy tradeoff.

Lemma 11 (Probe count under the threshold schedule) *Suppose the algorithm uses the threshold schedule*

$$a_m := \min\left\{1, \left(\frac{L_T}{m}\right)^{1/3}\right\}, \qquad m \ge 1.$$

On every sample path on which the combined confidence interval $\mathcal{C}_\mu(m)$ is nonempty for all $m \le T$, the number P_T of probe episodes initiated before horizon T satisfies

$$P_T \le 1 + \min\left\{T, \frac{L_T}{2a_T^2}\right\} \le 1 + L_T^{1/3} T^{2/3}.$$

In particular, for horizons satisfying $T \ge L_T$, we have

$$P_T \le 1 + \frac{L_T}{2a_T^2} = 1 + \frac{1}{2} L_T^{1/3} T^{2/3}.$$

L et P_m denote the number of completed probes in the shrinkage log at the start of episode boundary m. The first probe is allowed without any previous probe observation, and this accounts for the additive 1 in the bound.

Consider any later probe. Suppose the algorithm launches the $(p+1)$st probe at episode boundary m, where $p \ge 1$ probes have already been completed. Because the combined confidence set is an intersection that includes the probe certificate,

$$\mathcal{C}_\mu(m) \subseteq \mathcal{C}_{\mathrm{pr}}(m).$$

Since $\mathcal{C}_\mu(m)$ is nonempty, its radius is at most the radius of the probe interval. The probe radius after p completed probes is bounded by

$$r_{\mathrm{pr}}(m) \le \sqrt{\frac{L_T}{2p}},$$

and hence

$$r_\mu(m) \le r_{\mathrm{pr}}(m) \le \sqrt{\frac{L_T}{2p}}.$$

The algorithm launches a probe only if the current shrinkage radius exceeds the threshold:

$$r_\mu(m) > a_{m+1}.$$

Any probe counted in P_T is initiated before the period horizon T, and therefore $m+1 \le T$. Since a_m is nonincreasing,

$$a_{m+1} \ge a_T.$$

Thus

$$\sqrt{\frac{L_T}{2p}} > a_T,$$

so

$$p < \frac{L_T}{2a_T^2}.$$

If $P_T = K \ge 2$, applying this to the last probe gives

$$K - 1 < \frac{L_T}{2a_T^2},$$

and therefore

$$P_T \le 1 + \frac{L_T}{2a_T^2}.$$

Also, trivially, $P_T \le T$, because every probe lasts one period. Hence

$$P_T \le 1 + \min\left\{T, \frac{L_T}{2a_T^2}\right\}.$$

It remains to simplify the order. If $T \ge L_T$, then

$$a_T = \left(\frac{L_T}{T}\right)^{1/3},$$

so

$$\frac{L_T}{2a_T^2} = \frac{1}{2} L_T^{1/3} T^{2/3}.$$

If $T < L_T$, then the trivial bound gives

$$P_T \le T \le L_T^{1/3} T^{2/3}.$$

Combining the two cases yields

$$P_T \le 1 + L_T^{1/3} T^{2/3}.$$

This proves the lemma.

Lemma 12 (Cumulative shrinkage radius under the threshold schedule) *Along every sample path, each completed decision episode $m \in \mathcal{D}_T$ starts with*

$$r_m^- \le a_m.$$

Consequently,

$$\sum_{m \in \mathcal{D}_T} r_m^- \le \sum_{m=1}^{T} a_m \le 3 L_T^{1/3} T^{2/3}.$$

B y construction, episode m is a decision episode only if the algorithm did not probe at its start, which means precisely that $r_m^- \le a_m$. Since at most T episodes can be started before horizon T, it remains only to bound the sum of the threshold schedule. Using $a_m \le (L_T/m)^{1/3}$ and the integral estimate

$$\sum_{m=1}^{T} m^{-1/3} \le 1 + \int_1^T x^{-1/3}\,dx \le 3T^{2/3},$$

we obtain the result.

Lemma 13 (Sum of statistical widths) *Along every sample path,*

$$\sum_{m\in\mathcal{D}_T} B_{i_m}^{\mathrm{stat}}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\} \le 2c_{\mathrm{stat},T}\sqrt{MTL_{M,T}},$$

where i_m is the arm selected in decision episode m.

F or each arm i, let N_i^T be the number of completed decision episodes of that arm before horizon T. For the nth completed decision episode of arm i, the arm has been sampled before exactly when $n\ge 2$, and its statistical width is at most

$$c_{\mathrm{stat},T}\sqrt{\frac{L_{M,T}}{n-1}}.$$

Therefore

$$\sum_{m\in\mathcal{D}_T:i_m=i} B_{i_m}^{\mathrm{stat}}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\} \le c_{\mathrm{stat},T}\sqrt{L_{M,T}}\sum_{r=1}^{N_i^T-1} r^{-1/2} \le 2c_{\mathrm{stat},T}\sqrt{L_{M,T}N_i^T},$$

where the sum is zero if $N_i^T\le 1$. Summing over arms and using Cauchy–Schwarz gives

$$\sum_{i=1}^{M}\sqrt{N_i^T} \le \sqrt{M\sum_{i=1}^{M}N_i^T} \le \sqrt{MT},$$

which proves the bound.

Lemma 14 (Sum of implementation-bias widths) *On the event that all episodes initiated before horizon T have length at most $\bar\Lambda_T$,*

$$\sum_{m\in\mathcal{D}_T} B_{i_m}^{\mu,\mathrm{impl}}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\} \le L_{\mathrm{impl}}\bar\Lambda_T\log(eT)\sum_{m\in\mathcal{D}_T} r_m^-.$$

F ix an arm i, and enumerate its completed decision episodes before horizon T by

$$\ell_{i,1} < \ell_{i,2} < \cdots < \ell_{i,N_i^T}.$$

For the nth completed episode of this arm with $n\ge 2$, the implementation component computed before that episode satisfies, using $\Lambda_{\ell_{i,j}}\le\bar\Lambda_T$ and $\Lambda_{\ell_{i,j}}\ge 1$,

$$B_i^{\mu,\mathrm{impl}}(\ell_{i,n}-1) \le L_{\mathrm{impl}}\bar\Lambda_T\frac{\sum_{j=1}^{n-1} r_\mu(\ell_{i,j}-1)}{n-1}.$$

Thus

$$\sum_{n=2}^{N_i^T} B_i^{\mu,\mathrm{impl}}(\ell_{i,n}-1) \le L_{\mathrm{impl}}\bar\Lambda_T\sum_{n=2}^{N_i^T}\frac{1}{n-1}\sum_{j=1}^{n-1} r_\mu(\ell_{i,j}-1).$$

Interchanging sums gives

$$\sum_{n=2}^{N_i^T} B_i^{\mu,\mathrm{impl}}(\ell_{i,n}-1) \le L_{\mathrm{impl}}\bar\Lambda_T\log(eT)\sum_{j=1}^{N_i^T} r_\mu(\ell_{i,j}-1).$$

Summing over arms yields the claimed bound, since $r_\mu(\ell_{i,j}-1) = r_{\ell_{i,j}}^-$.

B.8. Proof of Theorem 1

L et

$$i^\star \in \arg\min_{1\le i\le M_T} g(s_i), \qquad S^\star_{M_T} := s_{i^\star},$$

and define the finite-grid learning regret

$$R_T^{\mathrm{grid}} := \mathbb{E}\left[\sum_{t=1}^{T} C_t^{\mathrm{alg}}\right] - T g(S^\star_{M_T}).$$

We first prove a grid-level bound and then add the deterministic discretization error. To simplify notation, write $M = M_T$ throughout the proof.

Let $\mathcal{D}_T$ be the set of completed decision episodes before horizon T, let P_T be the number of probe episodes initiated before horizon T, and write

$$r_m^- := r_\mu(m-1)$$

for the shrinkage radius available at the start of episode m.

For each completed decision episode $m \in \mathcal{D}_T$, let Γ_m be its total realized cost and i_m the selected arm. Let $\mathcal{G}_{m-1}$ denote the history up to the beginning of episode m, and let $\mathcal{E}_{m-1}$ be the prefix event that all confidence and concentration statements used in Lemma 9 hold up to that point. Since these prefix events are nested and Lemma 9 gives $\mathbb{P}(\mathcal{E}_T^c) \le 6T^{-4}$, we have

$$\mathbb{P}(\mathcal{E}_{m-1}^c) \le 6T^{-4} \qquad \text{for every } m \le T.$$

We decompose R_T^{grid} into completed decision episodes, probe episodes, and one possible boundary episode. The boundary contribution is bounded by a deterministic constant C_{bd} depending only on the model primitives. Probe episodes contribute at most

$$C_{\mathrm{pr}} := b\overline{D}$$

each, because a clean probe lasts one period and ends with zero inventory. Thus it remains to control

$$\sum_{m\in\mathcal{D}_T} \mathbb{E}[\Gamma_m - g(S^\star_{M_T})\Lambda_m].$$

For a fixed completed decision episode m, the pair (i_m, ν_{m-1}) is $\mathcal{G}_{m-1}$-measurable, and the episode starts from the stockout state. Therefore

$$\mathbb{E}[\Gamma_m - g(S^\star_{M_T})\Lambda_m \mid \mathcal{G}_{m-1}] = \mathbb{E}[\Lambda_m \mid \mathcal{G}_{m-1}]\big(g(s_{i_m};\nu_{m-1}) - g(S^\star_{M_T})\big).$$

By Lemma 2, the conditional mean cycle length is at most $1/p_0$.

On $\mathcal{E}_{m-1}$, if arm i_m has not been completed before, then

$$g(s_{i_m}) - g(S^\star_{M_T}) = \gamma(s_{i_m};\mu_\rho) - \gamma(S^\star_{M_T};\mu_\rho) \le 2\bar{c}_{\mathrm{per}},$$

so

$$g(s_{i_m};\nu_{m-1}) - g(S^\star_{M_T}) \le 2\bar{c}_{\mathrm{per}} + L_{\mathrm{impl}} r_m^-.$$

If the arm has been completed before, Lemma 9 gives

$$g(s_{i_m}) - g(S^\star_{M_T}) \le 2B_{i_m}(m-1),$$

and therefore

$$g(s_{i_m};\nu_{m-1}) - g(S^\star_{M_T}) \le 2B_{i_m}(m-1) + L_{\mathrm{impl}} r_m^-.$$

Hence, on $\mathcal{E}_{m-1}$,

$$\mathbb{E}[\Gamma_m - g(S^\star_{M_T})\Lambda_m \mid \mathcal{G}_{m-1}] \le \frac{2\bar{c}_{\mathrm{per}}}{p_0}\mathbf{1}\{N_{i_m}(m-1)=0\} + \frac{2B_{i_m}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\} + L_{\mathrm{impl}} r_m^-}{p_0}.$$

On the complement $\mathcal{E}^c_{m-1}$, Lemma 1 implies the crude bound

$$\mathbb{E}[\Gamma_m - g(S^\star_{M_T})\Lambda_m \mid \mathcal{G}_{m-1}] \le C_{\mathrm{ep}}$$

for a constant $C_{\mathrm{ep}} < \infty$ depending only on the primitives. Taking expectations and summing over completed decision episodes yields

$$\begin{aligned} R_T^{\mathrm{grid}} \le C_{\mathrm{bd}} + C_{\mathrm{pr}}\mathbb{E}[P_T] + \frac{2\bar{c}_{\mathrm{per}}}{p_0}M + \frac{2}{p_0}\mathbb{E}\left[\sum_{m\in\mathcal{D}_T} B_{i_m}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\}\mathbf{1}\{\mathcal{E}_{m-1}\}\right] \\ + \frac{L_{\mathrm{impl}}}{p_0}\mathbb{E}\left[\sum_{m\in\mathcal{D}_T} r_m^- \mathbf{1}\{\mathcal{E}_{m-1}\}\right] + C_{\mathrm{ep}}\sum_{m=1}^{T}\mathbb{P}(\mathcal{E}^c_{m-1}). \end{aligned} \tag{72}$$

Because $\mathbb{P}(\mathcal{E}^c_{m-1}) \le 6T^{-4}$ and there are at most T episode starts before horizon T, the last term is $O(T^{-1})$.

It remains to control the confidence radii. On every sample path on which $\mathcal{E}_{m-1}$ holds and $N_{i_m}(m-1)\ge 1$,

$$B_{i_m}(m-1) = B^{\mathrm{stat}}_{i_m}(m-1) + B^{\mu,\mathrm{eval}}_{i_m}(m-1) + B^{\mu,\mathrm{impl}}_{i_m}(m-1).$$

By Lemma 13,

$$\sum_{m\in\mathcal{D}_T} B^{\mathrm{stat}}_{i_m}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\}\mathbf{1}\{\mathcal{E}_{m-1}\} \le 2c_{\mathrm{stat},T}\sqrt{MTL_{M,T}}.$$

Next,

$$\sum_{m\in\mathcal{D}_T} B^{\mu,\mathrm{eval}}_{i_m}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\}\mathbf{1}\{\mathcal{E}_{m-1}\} \le L_\alpha \bar{S}\sum_{m\in\mathcal{D}_T} r_m^-.$$

Finally, whenever $\mathcal{E}_{m-1}$ holds, all previous episodes satisfy the bounded-length condition from Lemma 2; hence Lemma 14 gives

$$\sum_{m\in\mathcal{D}_T} B^{\mu,\mathrm{impl}}_{i_m}(m-1)\mathbf{1}\{N_{i_m}(m-1)\ge 1\}\mathbf{1}\{\mathcal{E}_{m-1}\} \le L_{\mathrm{impl}}\bar{\Lambda}_T\log(eT)\sum_{m\in\mathcal{D}_T} r_m^-.$$

Substituting these three bounds into (72) gives

$$\begin{aligned} R_T^{\mathrm{grid}} \le & C_{\mathrm{bd}} + \frac{2\bar{c}_{\mathrm{per}}}{p_0}M + \frac{4c_{\mathrm{stat},T}}{p_0}\sqrt{MTL_{M,T}} + C_{\mathrm{pr}}\mathbb{E}[P_T] \\ & + \left(\frac{2L_\alpha\bar{S} + L_{\mathrm{impl}}}{p_0} + \frac{2L_{\mathrm{impl}}\bar{\Lambda}_T\log(eT)}{p_0}\right)\mathbb{E}\left[\sum_{m\in\mathcal{D}_T} r_m^-\right] + O(T^{-1}). \end{aligned} \tag{73}$$

By Lemma 12,

$$\mathbb{E}\left[\sum_{m\in\mathcal{D}_T} r_m^-\right] \le 3L_T^{1/3}T^{2/3}.$$

By Lemma 11, together with the trivial bound $P_T \le T$ on the failure event,

$$\mathbb{E}[P_T] \le 1 + O(L_T^{1/3}T^{2/3}).$$

Since $\bar{\Lambda}_T$, L_T, and $L_{M,T}$ are logarithmic, the grid-level bound becomes

$$R_T^{\mathrm{grid}} = \widetilde{O}\Big(\sqrt{MT} + T^{2/3}\Big).$$

It remains to add the deterministic discretization error. By Lemma 10, for the uniform grid $\mathcal{S}_{M_T}$,

$$g(S^\star_{M_T}) - g(S^\star_{\mathrm{EV}}) \le L_S \frac{\bar{S}}{M_T}.$$

Therefore

$$R_T = R_T^{\mathrm{grid}} + T\{g(S^\star_{M_T}) - g(S^\star_{\mathrm{EV}})\} \le \widetilde{O}\Big(\sqrt{M_T T} + T^{2/3}\Big) + L_S \bar{S}\frac{T}{M_T}.$$

With $M_T = \lceil T^{1/3} \rceil$, both $\sqrt{M_T T}$ and T/M_T are of order $T^{2/3}$, up to constants. Hence

$$R_T = \widetilde{O}(T^{2/3}).$$

B.9. Proof of Theorem 2

T hroughout the proof write

$$P_+ := P_\Delta^+, \qquad P_- := P_\Delta^-.$$

Let $J_t \in \{P, L, H\}$ denote the action chosen in period t, corresponding to the base-stock levels S_P, S_L, S_H. Let

$$N_P(T) := \sum_{t=1}^{T} \mathbf{1}\{J_t = P\}$$

be the number of probe periods. Because the audited-reset subclass resets after every period, the observable transcript up to time t is

$$\mathcal{H}_t := \big(J_s, Y_s, Z_s\big)_{s=1}^{t}, \qquad \mathcal{H}_0 := \emptyset.$$

The conditional distribution of J_t is predictable from the pre-period transcript $\mathcal{H}_{t-1}$; for deterministic policies, J_t itself is predictable. If the policy randomizes, its randomization is taken to be independent of the primitive demand and shrinkage draws and is included in the induced transcript law. Let

$$\mathbb{P}_+^{\pi,T} \qquad \text{and} \qquad \mathbb{P}_-^{\pi,T}$$

denote the joint probability laws of the full transcript $\mathcal{H}_T$ induced by policy π under instances P_+ and P_-, respectively.

The one-period observation kernels are explicit. At $S_P = 1/2$,

$$O(S_P, 0) = (0,1), \quad O\left(S_P, \frac{1}{4}\right) = \left(\frac{1}{8}, 1\right), \quad O\left(S_P, \frac{1}{2}\right) = \left(\frac{1}{4}, 1\right), \quad O(S_P, 1) = \left(\frac{1}{2}, 1\right). \tag{74}$$

Thus the probe action reveals the shrinkage realization exactly. At $S_L = 2$,

$$O(S_L, 0) = (0,1), \qquad O\left(S_L, \frac{1}{4}\right) = \left(\frac{1}{2}, 1\right), \qquad O\left(S_L, \frac{1}{2}\right) = O(S_L, 1) = (1, 0). \tag{75}$$

The total probability of the aggregate event $\rho \in \{1/2, 1\}$ is $2/5$ under both P_+ and P_-. Hence the observation kernel under S_L is identical under the two instances. At $S_H = 4$,

$$O(S_H, 0) = (0,1), \qquad O\left(S_H, \frac{1}{4}\right) = O\left(S_H, \frac{1}{2}\right) = O(S_H, 1) = (1, 0). \tag{76}$$

The total probability of $\rho > 0$ is $4/5$ under both instances. Hence the observation kernel under S_H is also identical under P_+ and P_-. If K_σ^a denotes the one-period observation kernel under instance $\sigma \in \{+,-\}$ and action $a \in \{P, L, H\}$, then

$$K_+^L = K_-^L, \qquad K_+^H = K_-^H. \tag{77}$$

Since the policy rule is the same under both instances, the action-selection kernel contributes no KL divergence conditional on a common pre-period transcript. By the chain rule for KL divergence,

$$\mathrm{KL}(\mathbb{P}_+^{\pi,T}, \mathbb{P}_-^{\pi,T}) = \mathbb{E}_+\left[\sum_{t=1}^{T} \mathrm{KL}\big(K_+^{J_t}, K_-^{J_t}\big)\right]. \tag{78}$$

By (77), the operating actions contribute no KL divergence:

$$\mathrm{KL}(K_+^L, K_-^L) = \mathrm{KL}(K_+^H, K_-^H) = 0. \tag{79}$$

At S_P, the observation reveals ρ, so

$$\mathrm{KL}(K_+^P, K_-^P) = \mathrm{KL}(P_+^\rho, P_-^\rho). \tag{80}$$

The two shrinkage laws differ only in their masses on $1/2$ and 1. Hence

$$\mathrm{KL}(P_+^\rho, P_-^\rho) = \left(\frac{1}{5} + 2\Delta\right) \log \frac{\frac{1}{5} + 2\Delta}{\frac{1}{5} - 2\Delta} + \left(\frac{1}{5} - 2\Delta\right) \log \frac{\frac{1}{5} - 2\Delta}{\frac{1}{5} + 2\Delta}. \tag{81}$$

For $\Delta \le 1/20$, the Bernoulli KL bound gives

$$\mathrm{KL}(P_+^\rho, P_-^\rho) \le C_{\mathrm{KL}} \Delta^2 \tag{82}$$

for a numerical constant $C_{\mathrm{KL}} < \infty$. Combining (78)–(82) yields

$$\mathrm{KL}(\mathbb{P}_+^{\pi,T}, \mathbb{P}_-^{\pi,T}) \le C_{\mathrm{KL}} \Delta^2\, \mathbb{E}_+[N_P(T)]. \tag{83}$$

The same calculation in the reverse direction gives

$$\mathrm{KL}(\mathbb{P}_-^{\pi,T}, \mathbb{P}_+^{\pi,T}) \le C_{\mathrm{KL}} \Delta^2\, \mathbb{E}_-[N_P(T)]. \tag{84}$$

The one-period costs are

$$\begin{array}{c|cccc} & \rho = 0 & \rho = 1/4 & \rho = 1/2 & \rho = 1 \\ \hline S_P = 1/2 & 3 & 21/8 & 9/4 & 3/2 \\ S_L = 2 & 3 & 3/2 & 0 & 1 \\ S_H = 4 & 3 & 0 & 1 & 3. \end{array} \tag{85}$$

Using (85) and the definition of P_+,

$$\mathbb{E}_+[c(S_H, \rho) - c(S_L, \rho)] = 2\Delta. \tag{86}$$

Thus S_L is better than S_H under P_+, with gap 2Δ. Similarly,

$$\mathbb{E}_-[c(S_L, \rho) - c(S_H, \rho)] = 2\Delta, \tag{87}$$

so S_H is better than S_L under P_-, again with gap 2Δ.

The probe level has a constant regret price. Under P_+,

$$\mathbb{E}_+[c(S_P, \rho)] = \frac{12}{5} - \frac{3}{2}\Delta, \qquad \mathbb{E}_+[c(S_L, \rho)] = \frac{7}{5} + 2\Delta, \tag{88}$$

and therefore

$$\mathbb{E}_{+}[c(S_P,\rho)] - \mathbb{E}_{+}[c(S_L,\rho)] = 1 - \frac{7}{2}\Delta \geq \frac{1}{2} \tag{89}$$

for $\Delta \leq 1/7$. Under P_-,

$$\mathbb{E}_{-}[c(S_P,\rho)] - \mathbb{E}_{-}[c(S_H,\rho)] \geq \frac{1}{2} \tag{90}$$

for all sufficiently small Δ. Set

$$c_P := \frac{1}{2}. \tag{91}$$

Equations (86)–(91) imply the periodwise regret lower bounds

$$R_T^{\pi}(P_+) \geq c_P\,\mathbb{E}_{+}[N_P(T)] + 2\Delta \sum_{t=1}^{T} \mathbb{P}_{+}(J_t = H), \tag{92}$$

and

$$R_T^{\pi}(P_-) \geq c_P\,\mathbb{E}_{-}[N_P(T)] + 2\Delta \sum_{t=1}^{T} \mathbb{P}_{-}(J_t = L). \tag{93}$$

Define the two-point average regret and average number of probes by

$$\bar{R}_T := \frac{1}{2}\{R_T^{\pi}(P_+) + R_T^{\pi}(P_-)\}, \qquad \bar{N}_P := \frac{1}{2}\{\mathbb{E}_{+}[N_P(T)] + \mathbb{E}_{-}[N_P(T)]\}. \tag{94}$$

If

$$\bar{N}_P \geq \frac{1}{64 C_{\mathrm{KL}} \Delta^2}, \tag{95}$$

then (92)–(94) imply

$$\bar{R}_T \geq c_P \bar{N}_P \geq \frac{c_P}{64 C_{\mathrm{KL}}} \Delta^{-2}. \tag{96}$$

It remains to consider the complementary case

$$\bar{N}_P < \frac{1}{64 C_{\mathrm{KL}} \Delta^2}. \tag{97}$$

Then

$$\mathbb{E}_{+}[N_P(T)] < \frac{1}{32 C_{\mathrm{KL}} \Delta^2}. \tag{98}$$

Combining (83) and (98) gives

$$\mathrm{KL}(\mathbb{P}_{+}^{\pi,T}, \mathbb{P}_{-}^{\pi,T}) < \frac{1}{32}. \tag{99}$$

Pinsker's inequality implies

$$\mathrm{TV}(\mathbb{P}_{+}^{\pi,T}, \mathbb{P}_{-}^{\pi,T}) < \frac{1}{8}. \tag{100}$$

By data processing, the same total-variation bound holds for the distribution of the pre-period history before every period t.

For each period t, define

$$n_t := \frac{1}{2}\left[\mathbb{P}_{+}(J_t \in \{L,H\}) + \mathbb{P}_{-}(J_t \in \{L,H\})\right], \tag{101}$$

$$e_t := \frac{1}{2}\left[\mathbb{P}_{+}(J_t = H) + \mathbb{P}_{-}(J_t = L)\right], \qquad c_t := \frac{1}{2}\left[\mathbb{P}_{+}(J_t = L) + \mathbb{P}_{-}(J_t = H)\right].$$

Then $n_t = c_t + e_t$. Let

$$g_t(h) := \mathbb{P}_{\pi}(J_t = L \mid \mathcal{H}_{t-1} = h) - \mathbb{P}_{\pi}(J_t = H \mid \mathcal{H}_{t-1} = h)$$

be the policy's predictable signed operating decision at period t. For a deterministic policy, this reduces to $\mathbf{1}\{J_t = L\} - \mathbf{1}\{J_t = H\}$. In all cases, $|g_t(h)| \le 1$. Let

$$d_t := \sup_{\|f\|_\infty \le 1} \left|\mathbb{E}_+[f(\mathcal{H}_{t-1})] - \mathbb{E}_-[f(\mathcal{H}_{t-1})]\right|.$$

By (100) and data processing,

$$d_t \le \frac{1}{4}. \tag{102}$$

Moreover,

$$c_t - e_t = \frac{1}{2}\left(\mathbb{E}_+[g_t(\mathcal{H}_{t-1})] - \mathbb{E}_-[g_t(\mathcal{H}_{t-1})]\right) \le \frac{1}{2} d_t.$$

Using $n_t = c_t + e_t$, we get

$$e_t = \frac{1}{2}\{n_t - (c_t - e_t)\} \ge \frac{1}{2} n_t - \frac{1}{4} d_t \ge \frac{1}{2} n_t - \frac{1}{16}. \tag{103}$$

Summing (103) over $t = 1, \ldots, T$ gives

$$\sum_{t=1}^{T} e_t \ge \frac{1}{2} \sum_{t=1}^{T} n_t - \frac{T}{16}. \tag{104}$$

Since

$$\sum_{t=1}^{T} n_t = T - \bar{N}_P,$$

we have

$$\sum_{t=1}^{T} e_t \ge \frac{1}{2}(T - \bar{N}_P) - \frac{T}{16}. \tag{105}$$

For $\Delta = \Delta_T = aT^{-1/3}$, condition (97) implies $\bar{N}_P \le T/4$ for all sufficiently large T. Hence (105) gives

$$\sum_{t=1}^{T} e_t \ge \frac{5T}{16} \tag{106}$$

for all sufficiently large T.

Averaging (92) and (93), and using (106), yields

$$\bar{R}_T \ge 2\Delta \sum_{t=1}^{T} e_t \ge \frac{5}{8} T\Delta. \tag{107}$$

Equations (96) and (107) imply that, for every nonanticipative policy π,

$$\sup_{\sigma \in \{+,-\}} R_T^\pi(P_\Delta^\sigma) \ge \bar{R}_T \ge c \min\{\Delta^{-2}, T\Delta\}, \tag{108}$$

for a numerical constant $c > 0$, whenever $\Delta \le \min\{1/20, 1/7\}$ and T is sufficiently large.

Choose

$$\Delta_T := aT^{-1/3}$$

with $a > 0$ small enough that

$$\Delta_T \le \min\{1/20, 1/7\}$$

for all $T \ge T_0$. Then

$$\Delta_T^{-2} \asymp T^{2/3}, \qquad T\Delta_T \asymp T^{2/3}.$$

Substituting $\Delta = \Delta_T$ into (108) gives

$$\sup_{\sigma \in \{+,-\}} R_T^\pi(P_{\Delta_T}^\sigma) \ge cT^{2/3}.$$

Taking the infimum over all nonanticipative policies proves Theorem 2.